THE OKLO FILE

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James Edmund Carpenter II

Earth's Natural Nuclear Reactors

HOW TO READ THIS FILE

Hello. Thank you for picking this up. Before we go anywhere, let me tell you how this file works, because it is not written like a textbook and it is not written like a thriller, though it contains pieces of both.

This is a file about a real place — a uranium mine in the country of Gabon, in west-central Africa — where, about two billion years ago, nature built nuclear reactors and ran them for hundreds of thousands of years. No person designed them. No person ran them. They ran on their own, in the dark, deep underground, long before there were trees, long before there were fish, long before there was anything with a spine. And then they shut down, and the ground held their waste in place for two billion years, until human beings dug into it by accident.

Everything in this file comes from evidence: scientific papers, government reports, corporate filings, court records, conference proceedings, museum records, and newspapers. Every time a number or a story appears, it has a pedigree — a note about where it comes from — and when the sources disagree with each other, this file tells you they disagree and shows you the range. Where a claim has no evidence, this file says so, plainly, and documents the claim anyway, because even a false claim is evidence of something — usually of what people wanted to be true.

Three rules governed the writing:

1. **No adopted theories.** This file does not join schools of thought. It lays out what the evidence shows, what different

researchers concluded, and what remains unknown.

2. **Name names.** The people who did this work — geologists, chemists, physicists, miners, lawyers, activists, and yes, the people who promoted wild ideas — are named, because history happens to people, not to abstracts.

3. **No blame.** Some of what you will read involves sickness, death, and long legal struggles. This file does not prosecute anyone. It presents the evidence as it stands — what the company said, what the workers said, what the courts decided, what the science shows — and leaves judgment to you.

One more thing before we start. This file is long, and it is meant to be read from the beginning, like a letter from a friend who happened to fall down a very deep hole and is now writing to tell you everything they found down there. You do not need any background. Every technical idea is explained when it arrives, in plain language, with its full detail. If you read patiently, by the end you will understand this subject better than almost anyone you know.

Let us begin where the story begins: with a missing two hundred kilograms of uranium, in the summer of 1972.

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PROLOGUE: THE MISSING TWO HUNDRED KILOGRAMS

In June of 1972, at a uranium enrichment plant — the kind of factory that increases the share of the fissile flavor — called Pierrelatte, in the south of France, a chemist ran a routine test and got a routine-looking result that was not routine at all. It was, in fact, impossible — and he knew it.

Here is what you need to know to feel the weight of that moment. Natural uranium — the heavy metal that fuels nuclear reactors, as dense as gold: a lump the size of a soda can weighs about three kilos (six and a half pounds) — is, everywhere on Earth, in every rock and every meteorite and even in the dust the Apollo astronauts brought back from the Moon, has exactly the same recipe: 99.27 percent uranium-238, and 0.72 percent uranium-235, with a tiny trace of uranium-234. (There are two ways to quote that famous number: by weight, uranium-235 is 0.7202 percent; counting individual atoms, it is 0.7110 percent. Different books use different conventions, which has confused more than one reader. This file will always tell you which convention a number uses.) That ratio is a fingerprint. It never varies, because the two isotopes are made by the same cosmic processes and decay at fixed, known rates. Uranium-235 is the rarer one, and it is the one that matters, because uranium-235 is the kind of uranium that can sustain a nuclear chain reaction. The other kind mostly cannot.

So when the analysts at Pierrelatte measured the uranium-235 content of a batch of uranium hexafluoride gas that

had come from Gabon, and got 0.7171 percent by weight instead of 0.7202 — a deficit of about three thousandths of one percent — they were not looking at a rounding error — three thousandths of one percent is about the size of a tenth of an inch on a football field. They were looking at uranium that had been touched by something. The fingerprint had been altered.

The first suspicion was the worst one: theft. Missing uranium-235 is bomb material. The deficit, spread over the hundreds of tons of ore the mine had shipped since 1970, added up to roughly two hundred kilograms — about 440 pounds — of the fissile isotope — enough, people said at the time, for six atomic bombs. One account from CEA veteran Jean-François Dozol adds a Cold War detail that reads like a novel: the anomalous batch was uranium that France was preparing to ship to the Soviet Union for enrichment. Imagine the espionage calculations that would have followed a deficit found in *that* pipeline.

For weeks, the CEA's scientists chased the anomaly through every stage of the production chain — from the enrichment plant back through the conversion plants at Malvési and Gueugnon, across the ocean to the mill — the factory where the ore is crushed and dissolved — at Mounana in Gabon, and finally to the open pit at a place called Oklo, two kilometers south of the town of Mounana. They tested the obvious explanation first: that the uranium had been contaminated with recycled enrichment tailings. They eliminated it in a beautiful way — recycled uranium contains uranium-236, a telltale isotope left over from reactor use, and there was no uranium-236 in the samples.

Whatever had happened had happened in the ground.

What they found was not a thief. It was a reactor.

The uranium was depleted because some of its uranium-235 had been *fissioned* — actually split, atom by atom, in a sustained chain reaction — roughly two billion years earlier. The missing uranium had not been stolen. It had been burned as fuel, by a reactor that nobody built, buried deep beneath what was then a planet inhabited only by single-celled life.

The announcement came on September 25, 1972. Francis Perrin, a former head of the French atomic energy commission, presented two short papers at the Academy of Sciences in Paris — one on the isotopic anomalies, one on the natural chain reaction — papers whose author lists included Michèle Neuilly, Jean Bussac, Claude Fréjacques, Guy Nief, Georges Vendryes, and Jacques Yvon. The following day, a French delegate named Giraud stood up at the General Conference of the International Atomic Energy Agency in Vienna and told the world, associating the Gabonese delegation with the news, since the discovery belonged to their country's ground. The New York Times carried the story on September 26, 1972. The scientific world was stunned, and, as we will see later, a few people who had predicted exactly this possibility allowed themselves a quiet, enormous satisfaction.

Mining paused at Oklo while an international effort — the Franceville Project, led on the French side by Roger Naudet — turned a working uranium mine into the most unusual laboratory on Earth.

This file is the complete record of what has been learned since. It is organized as a walk: first how the discovery happened and who really did what; then the ground itself; then, step by step, exactly how these reactors worked — every mechanism, every number, every debate; then the evidence, isotope by isotope, including the astonishing use of Oklo to test whether the laws of physics have changed; then the waste lesson that nuclear programs around the world still cite; then the money, the company, and the weapons connection; then the people — the scientists and the miners, including the long struggle over the health of the men who dug this ore; then the claims that the reactor was not natural at all, examined one by one as evidence; and finally the circle that closes the story: how Oklo taught humanity how to think about burying its own waste, and how a startup named after the site is trying to build reactors again. Along the way, the file keeps a running list of every error and inconsistency found in the published record, because honest work includes admitting where the record itself has stumbled.

Welcome to the file. Keep reading.

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A ONE-PAGE PRIMER, BEFORE THE DEEP WATER

The prologue threw you a few words it promised to honor later. Here, in one page, are the foundations — read this once and the whole file will hold your hand the rest of the way.

Uranium. A metal — a heavy, gray, ordinary-looking one, except for what it hides. It is the heaviest element found in quantity in the Earth's crust, nearly as dense as gold, and a few of its atoms are unstable in the most useful way known to physics: they can be made to split and release energy. That is why it is the fuel of nuclear reactors and bombs. In ore, before anyone refines it, it is a yellow-brown stain in rock — the yellow crust at Oklo was uranium oxide.

Atoms and elements. Everything solid is made of atoms, and an element is one *kind* of atom — iron is one element, gold another, uranium a third. Atoms are absurdly small: a single grain of sand contains more atoms than there are grains of sand on a beach.

Isotopes. Many elements come in flavors — the same kind of atom with slightly different weights. The flavors of uranium are 238 and 235. They are chemically identical, but 235 is the one whose nucleus can sustain a fire. The prologue gave you the recipe.

Neutrons. At the center of every atom sits its nucleus, and one of the particles that can live there is the neutron — heavy, uncharged, and perfect as a projectile. Fire a neutron into the right nucleus and the nucleus can split.

Fission. That splitting is fission: the atom breaks in two, releases energy, and throws off two or three fresh neutrons. It is, in effect, a fire that produces its own sparks.

Chain reaction. If at least one spark from each split sets off another split, the fire sustains itself — that is a chain reaction, and a pile of material that achieves it is said to be *critical*. If each split produces less than one new split, the fire dies; more, and it grows. A reactor is simply a pile arranged so the fire holds steady — and the Oklo reactors were piles of wet rock that held steady on their own for hundreds of thousands of years.

Radioactivity and half-life. Unstable atoms also split or transform *by themselves*, on a fixed schedule no force can hurry — this is radioactivity, a slow-motion fire that needs no air. Each flavor keeps its own clock: its half-life, the time in which half of any batch transforms. Uranium-235's clock runs for 704 million years. This is why the uranium of two billion years ago was richer in 235 — the fast-ticking flavor was still abundant — and why nothing on Earth can go critical today: too much of it has ticked away.

That is all the physics you need. The rest of the file will introduce every other idea where it is used, explain it in place, and give you something to hold whenever a number gets hard to picture.

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PART I — THE DISCOVERY

1. Before 1972: the prediction everybody ignored

The Oklo reactors were discovered in 1972, but they were *predicted* — twice — decades earlier, and both times the prediction was more or less ignored. This matters, because it tells you something about how science actually works: the right answer can be sitting in plain sight for twenty years while everyone assumes the subject is settled.

The first prediction came in 1953, from two American scientists, George Wetherill of UCLA and Mark Inghram of the University of Chicago, in a paper on spontaneous fission. They noted that some uranium deposits were a fair fraction of the way toward being self-sustaining reactors, and they calculated that if you rolled the clock back far enough — to a time when uranium-235 was more abundant — a rich enough deposit could actually go critical on its own. It was a remark, a speculation, not a full analysis.

The second prediction came three years later, in 1956, and it was rigorous. A Japanese chemist named Paul Kuroda — Paul Kazuo Kuroda, then at the University of Arkansas — published two short papers in the Journal of Chemical Physics laying out, for the first time, the exact conditions a uranium deposit would need to become a natural nuclear reactor. Kuroda had walked through the ruins of Hiroshima in August 1945 as a young technical expert sent to survey the damage, and the sight had never left him; he spent the rest of his life thinking about where nuclear energy came from and where it might lead. His 1956

conditions were, in plain language:

1. The uranium deposit had to be big enough — bigger than about two-thirds of a meter across, because that is roughly how far fission neutrons travel in rock, and a smaller lump would leak its neutrons away faster than the chain reaction could build.
2. The uranium had to be rich — because two billion years ago, nature's uranium contained about 3.5 percent uranium-235, not today's 0.72 percent. (This is not an assumption; it is simple decay arithmetic. Uranium-235 decays about six times faster than uranium-238, so the farther back in time you go, the more of it there used to be. At the formation of the solar system it was around 17 percent. It crossed downward through the 3 percent line between about 1.8 and 1.7 billion years ago, which fixes the window during which natural reactors were possible — and closes it forever once passed.)
3. There had to be a *moderator* — something to slow the neutrons down — and there had to be almost nothing that *absorbs* neutrons (what engineers call poisons). Water is a fine moderator. And the rocks around Oklo were unusually poor in neutron-absorbing elements, for reasons we will come to.

Kuroda also had a personal story worth telling: his papers were short notes in the Journal of Chemical Physics, which at the time would publish brief communications without full peer review, and he later recalled that this was the only reason they saw print at all — subject had been, in his words, unpopular for

sixteen years. Some colleagues dismissed him. He went from eccentric to prophet, he said, when Oklo was found. He lived until 2001, long enough to see his conditions verified isotope by isotope, and in 1982 the University of Arkansas held an event, the Kuroda Symposium, where the scientists who confirmed his work gathered to honor him; and years earlier he had predicted that the long-extinct isotope plutonium-244 once existed in the early solar system — a prediction vindicated in 1965 when his own laboratory detected its fossil fission xenon in the Pasamonte meteorite. His two 1956 papers were later selected for a volume of the eighty-five benchmark papers in nuclear chemistry.

There is an even earlier ancestor worth naming: in 1940, the Soviet physicists Yakov Zeldovich and Yulii Khariton — who would soon direct the Soviet atomic bomb program — analyzed whether a natural chain reaction could ever have occurred. Their remark predates Kuroda by sixteen years, and a French researcher named Jacques Noetzlin had searched volcanic rocks for traces of ancient fission in the years before that. And as far back as 1909, the radiochemist Frederick Soddy — who won the Nobel Prize for his work on radioactivity — gave a series of popular lectures in which he speculated that past civilizations might have known atomic energy and destroyed themselves with it. That speculation, as we will see in Part IX, became the seed of an entire genre of claims.

But in 1956, when Kuroda published, nobody had ever found such a reactor, and most geologists considered the whole idea a curiosity. Sixteen years later, the curiosity was standing in an open pit in Gabon.

2. How Gabon came to have a uranium mine at all

The discovery was only possible because France needed uranium, and France needed uranium because of decisions made in the 1940s. In April 1946, the French state declared all radioactive substances across its overseas territories to be state property. The Commissariat à l'Énergie Atomique — the CEA, France's atomic energy commission — began a systematic search of equatorial Africa, setting up a prospecting base called the *base Maba* at Franceville, in what was then French Equatorial Africa. It took ten years. In December 1956, using the new scintillometer counters — handheld devices that turn radiation into measurable flashes — that had just transformed uranium prospecting, CEA geologists confirmed a major deposit at a place called Mounana — a local name that, charmingly, means "the holes," after the drill holes the prospectors left behind.

A mining company was chartered on February 21, 1958: the Compagnie des Mines d'Uranium de Franceville, COMUF, owned by a consortium of French mining companies with the CEA behind it. Gabon became independent in August 1960, and COMUF began producing uranium in 1961 — first from an open pit at Mounana, with the ore shipped raw to France for processing, and later from its own mill at Mounana, which entered service on March 27, 1961. The yellowcake — the crude uranium powder, named for its color, that a mill actually ships — traveled by truck to the railhead at Mbinda, through the Congo port of Pointe-Noire, and across the sea to France. For many

years, France was the only customer.

Other deposits were found in the years that followed: Mikouloungou in 1965, Boyindzi in 1967, and — the one this file is about — Oklo in 1968, two kilometers south of Mounana. Mining at Oklo began around 1970. The district's deposits were shallow, rich, and, as it turned out, hiding sixteen nuclear reactors.

None of this was exotic at the time. Uranium was a quiet, strategic commodity; Gabon was a reliable supplier; and the men doing the work were industrial miners doing industrial work, with no idea what lay in the northern end of the Oklo pit.

3. The discovery itself: who did what, exactly

Because the discovery of the century happened inside a secretive, military-linked industry, the credit has been tangled from the start — and untangling it is part of this file's job. Here is the chain, name by name, with the surviving disagreements noted honestly.

The observation — Henri Bouzigues, at Pierrelatte, June 1972. The mass spectroscopist at the enrichment plant noticed the slight uranium-235 deficit in a routine analysis. The sources give four different dates for this moment — May 1972 (George Cowan's contemporary Los Alamos report), June 2 (a later Chinese-language summary), June 14, 1972 (Dozol's first-person account, the most precise), and simply "June" (the official IAEA record). We present all four, with Dozol's eyewitness date weighted most heavily, and note that the discrepancy itself tells you something: even the founding moment of this story was not

written down as carefully as it deserved.

The confirmation — the Cadarache team, early July 1972. Ore samples from Oklo were analyzed at the CEA's laboratory at Cadarache. Jean-Claude Nimal later recalled running the leftover material through a spark-source mass spectrometer and seeing something he had never seen in any natural material: the rare-earth element samarium had an impossible isotopic makeup — one isotope, samarium-149, was drastically depleted while its neighbors were not. Natural samarium always has the same recipe; this sample did not. Meanwhile Michèle Neuilly's team chemically separated and measured the rare earths and concluded, at first, that the chemistry pointed to a nuclear explosion — her initial hypothesis, recorded by her colleague Etienne Roth, was corrected within weeks to the far stranger truth: a sustained chain reaction, not a bomb.

The explanation — Guy Nief at Saclay, July 19, 1972 onward. At a CEA meeting on July 19, Roth suggested searching for the rare earths as the fingerprint of fission. By August 7, Nief had worked out, from the neodymium isotopes, exactly how the deposit had behaved as a reactor — and even estimated its power. The chain was closed in less than three months from the first observation, a pace that the Soviet physicist Yuri Petrov later singled out for praise.

The announcement — September 25, 1972. Francis Perrin, by then the *former* head of the CEA, presented the two Academy notes. The CEA's serving head, André Giraud, is named by later reviews as the announcing authority; Perrin presented.

This distinction matters, because popular accounts almost universally say Perrin "discovered" the phenomenon, and the IAEA's own outreach materials have dramatized him standing at the plant making the measurement — which he did not. The analysts were Bouzigues at Pierrelatte and the Cadarache team; the presenters were Perrin and Giraud. The lead investigator of the following years was Roger Naudet. The lead geologist was François Gauthier-Lafaye. Everyone else, including this file's sources, has spent fifty years clarifying these roles, and this file presents the clarified version.

The publications. Three notes appeared in the Comptes Rendus of the Academy in 1972: Bodu, Bouzigues, Morin and Pfiffelmann (volume 275, page 1731); Neuilly, Bussac, Fréjacques, Nief, Vendryes and Yvon (volume 275, page 1847); and Baudin, Blain, Hagemann, Kremer, Lucas, Merlivat, Molina, Nief, Prost-Marechal, Regnaud and Roth (volume 275, page 2291, presented by Perrin). Every modern reference list confirms these citations, and it is worth knowing that one frequently-cited review paper (Davis and colleagues, 2014) misprinted the volume number as 237 in its reference list — a small error that has propagated into other papers, which is why this file tracks such things.

The instrument. One small invention from the discovery years deserves recording: the "Oklometer," a scintillation device built so the mine could measure uranium-235 enrichment on the spot, without shipping samples to a mass spectrometer. Two were made — one at Mounana, one at Saclay — agreeing with the big machines to within about a percent and a half. It was the

first instrument ever built for a specific fossil reactor, and it retired with the mines.

The reception. The first press coverage was oddly small. The New York Times ran Walter Sullivan's story on September 26, 1972, and New Scientist followed in November; in Japan, only the Mainichi and the Nikkei carried the news at all, and in small items. The first Japanese scientist's pilgrimage to the site came five years later, in 1977. The world's attention was elsewhere — Vietnam, Watergate, the oil crisis brewing — and a two-billion-year-old reactor in an African pit was, for most readers, a curiosity to clip and forget. The scientists did not forget. Within three years the subject had its own international working group, and within a generation its own anniversary symposiums.

4. The two-year shutdown and the Franceville Project

Once the reactors were recognized, the CEA and the Gabonese government agreed to suspend mining at the anomalous zone so it could be studied. The Oklo deposit sits two kilometers south of Mounana, and the northern part of it — where the depleted ore was — was fenced off from the working pit. For roughly two years, the science came first.

In June 1975, the International Atomic Energy Agency, the CEA, and the government of Gabon convened an international symposium in Libreville, the capital, followed by field trips to the mine. Seventy investigators from about twenty countries sat on benches at the edge of Reactor Zone 2 — which had been

marked out with pegs and strings like an archaeological dig — while the chief geologist of COMUF, J. P. Bindeman, lectured on the geology from a palm-frond-decorated rostrum. The proceedings, published as *The Oklo Phenomenon* (IAEA STI/PUB/405, a 646-page volume), remain the foundation of the field. Two years later, in December 1977, a follow-up technical committee meeting in Paris (STI/PUB/475, 754 pages) brought sixty-six specialists from twelve countries and settled many of the open questions — and, significantly, reframed the whole subject: the 1977 meeting was where Oklo began to be studied explicitly as *a natural analogue for the safety of buried nuclear waste*. That reframing would drive four decades of international research.

By then the first six reactor zones were fully characterized, and more were being found — zones 7, 8 and 9 in 1978 at the southern end of the pit; deeper zones (10, 13, 16) and the Okélobondo zone (OK84) found by drilling in 1982; and, also in 1982, the exploration that would lead to the Bangombé zone. The definitive modern count, from Gauthier-Lafaye's own 2002 synthesis, is sixteen reactor zones in two deposits: fifteen at Oklo-Okélobondo and one at Bangombé, thirty kilometers south. (The literature also contains counts of fourteen, fifteen, and seventeen; the differences come down to whether the deep Okélobondo zone is counted with Oklo, and whether two numbered candidates that were never confirmed are included. This file uses sixteen, and tells you why.)

5. The errors we corrected along the way

Because this file promises to show its work, here, early, are the main corrections and inconsistencies uncovered in the published record during the research — each is flagged again at the place in the story where it matters:

- The IAEA's popular features credit Perrin with the discovery analysis; the primary French record shows he presented, and did not perform, the work. [Corrected in Part I.]
- The volume of the 1972 Academy notes is 275; a major review printed 237. [Fact-check file.]
- The 1975 symposium was held in Libreville; Naudet's preview article said Franceville. The venue discrepancy exists inside the primary record itself. [Noted.]
- The natural uranium-235 abundance is 0.7202 percent *by weight* or 0.7110 percent *by atom*; sources mix these freely. This file states which is which, every time. [Standing rule.]
- Several popular and even institutional sources give wrong ages or enrichments — "1.7 billion years at 3.1 percent," "1.8 billion years at 1.8 percent" — each internally inconsistent; the canonical values are 1.95 billion years and about 3.5 percent. [Fact-check file.]
- One widely-cited review's reference list contains systematic errors (wrong volume, a wrong year, a wrong co-author); the correct citations are given in this file's bibliography. [Fact-check file.]

- A parliamentary report says the Bangombé reactor ran for "500 years"; the primary literature says on the order of a hundred thousand years or more. The parliamentary figure is wrong by a factor of about two hundred. [Fact-check file.]
- Wikipedia's per-reactor inventory table overstates fission-product masses by a factor of about 450 and mislabels a whole-field energy figure as per-reactor; corrected figures are derived and given in Part III. [Corrected.]

There are more — a running list is kept in Appendix B — and they are presented not to embarrass anyone, but because an honest file shows where the record has stumbled.

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PART II — THE SITE

1. Gabon, and the Franceville Basin

Gabon sits on the Atlantic coast of equatorial Africa, straddling the equator, about eighty-five percent of it covered in rainforest. It is a small country in population — around two and a half million people — and rich in what lies under the forest: oil, manganese, iron, and, for forty years, uranium. The uranium country is in the southeast, in a province called Haut-Ogooué, where the towns of Franceville, Moanda, and Mounana sit in a landscape of red-earth roads and equatorial green.

The geology that matters here is the Franceville Basin: a lens of ancient sedimentary rock, roughly thirty thousand square kilometers, laid down on top of a much older crystalline basement about 2.7 billion years old. The sedimentary pile — geologists call it the Francevillian, after the town — is divided, from bottom to top, into formations labeled FA through FE. Think of it as a layer cake laid down between about 2.15 and 2.05 billion years ago:

- **FA** is the basal formation, 100 to 1000 meters of sandstones and conglomerates (sandstone with pebbles cemented in) — the deposits of rivers and then shallow seas, coarse and poorly sorted at the bottom, becoming finer and better sorted upward. At Oklo, the uppermost ten meters or so of the FA is a particular layer, about ten meters thick, that geologists call C1.

- **FB** is a thick package of dark shales and mudstones (compacted mud), some of them extraordinarily rich in organic matter — up to fifteen percent organic carbon. These are the remains of ancient microbial life, buried and cooked.
- **FC, FD, FE** above are cherts, dolomites, more shales, and volcanic tuffs (chert is flint; dolomite a magnesium-rich limestone; tuff is compacted volcanic ash); a date from the FD tuffs gives 2083 million years.

The uranium deposits of the district — Mounana, Oklo-Okélobondo, Boyindzi, Mikouloungou, and little Bangombé — sit along the northwest edge of this basin, strung out along the boundary of an old basement ridge (the ancient rock floor beneath the sedimentary layers). They are all in or near the C1 layer at the top of the FA, right against the FB shales. And here is the first great clue: the deposits with reactors (Oklo-Okélobondo and Bangombé) are precisely the ones where the uranium-bearing fluids met abundant organic matter from those black shales.

2. How the ore formed — the trap that made reactors possible

Here is the sequence, worked out over decades by the Strasbourg school — François Gauthier-Lafaye and Francis Weber above all — and confirmed by the international programs. Take it slowly; it is one of the most beautiful causal chains in geology, and every step is needed.

Step one — oxygen. For the first two billion years of Earth history, the atmosphere had almost no free oxygen. Uranium in

rocks exists mostly in a reduced form that does not dissolve in water. Then, around 2.4 billion years ago, photosynthetic microbes — cyanobacteria, the ancestors of today's blue-green algae — began pumping oxygen into the air and water. This event, the Great Oxidation Event, is one of the great divides of Earth history. And it had a specific chemical consequence: oxygen converts uranium from a water-hating form to a water-loving form. For the first time, uranium could dissolve into groundwater, flow, and be transported. (A small honest admission from the popular literature: uranium does not dissolve in *pure* distilled water — it needs the oxygen to change its chemistry, and real ground water with its dissolved content does the job nicely.)

Step two — the trap. The Franceville basin was buried kilometers deep, heated to around 140 to 200 degrees Celsius. Organic matter in the FB shales was cooked into kerogen (the waxy organic matter that heat and pressure turn into oil), then into bitumen and oil, which migrated. Where the migrating hydrocarbons met oxygenated, uranium-bearing waters moving through the C1 sandstone, the chemistry flipped: the hydrocarbons consumed the oxygen, uranium came out of solution and precipitated as pitchblende and uraninite — two names for uranium oxide ore, the black mineral that built the cores. The same organic matter did something equally important: it acted as a *filter*, keeping neutron-absorbing elements out of the richest ore. The deposits that failed to make reactors — Mounana, Boyindzi, Mikouloungou — are the ones where the system also concentrated vanadium, in quantities up

to 1.5 percent of the rock, and vanadium is a neutron poison. Oklo's ore carries less than a tenth as much. Life made the trap; life also armed it.

Step three — burial and the clock. The mineralization is dated at 2050 million years, plus or minus thirty. Uranium-235, as you know from Part I, was then about 3.5 percent of the total — about the same enrichment as the fuel in a modern power reactor, a coincidence Kuroda had already pointed out. At burial depths of two to four kilometers, with hot groundwater circulating, the conditions for criticality assembled themselves.

The result, in cross-section, is striking: lens-shaped ore bodies, a few hundred cubic meters each, of which the best are twenty to sixty percent uranium — locally up to eighty or ninety percent — enclosed in clay, sitting like dark loaves in the sandstone. Sixteen of these lenses went critical.

3. The deposits, one by one

Mounana — the first and the namesake district. Discovered 1956, mined from 1961: an open pit a hundred meters deep and about two hundred meters across, later underground workings with shafts named Simone and Émile. The deposit here is uranium-vanadium, and it produced the district's vanadium minerals, several of which — francevillite, mounanaite, chervétite, and others — were first described from this ground and named for it. The quarry was enlarged in the early 1970s; by 1975 the main Mounana deposit was exhausted, and the town's production shifted to the newer mines.

Oklo-Okélobondo — the reactor ground. The Oklo deposit, 900 meters north-south by 600 east-west, with a mineralized layer six to eight meters thick, was worked by open pit from 1970 and by underground galleries into the 1990s. The reactors sat in the northern part of the pit — zones 1 through 6 — with zones 7 through 9 a couple of hundred meters south, and the deep Okélobondo zone (OK84) about a kilometer and a half to two kilometers south, at 300 to 450 meters depth, reached only by drilling. Oklo's total output was roughly fifteen thousand tons of uranium, more than half the district's.

Bangombé — the small one that survived. A shallow deposit, 1.5 square kilometers, with the reactor zone barely twelve meters below the surface — about the depth of a four-story building, so shallow it could never be mined. It was located in 1982 during exploration, and its reactor was confirmed by drilling in 1986. Because it was too shallow to exploit economically, Bangombé is the one reactor that remains in the ground today, the last surviving natural reactor on Earth — a point we will return to in Part X, because the scientific community fought a real campaign to keep it.

Boyindzi — found 1967, mined 1980 to 1991, about 2,700 tons of uranium.

Mikouloungou — found 1965, mined at the very end, 1997 to 1999, about 1,134 tons.

When the district closed in June 1999, the mines were flooded by the rising water table and the rains; the galleries that once held the reactors are underwater now. The open pits are quiet lakes of red water.

4. The mining record, plainly

Because the numbers in different official records do not match, this file gives you the full set, with the authorities named, rather than silently picking one.

The OECD-IAEA *Red Book* — the standard international reference, produced by the club of rich-country governments together with the atomic agency — gives Gabon's lifetime production as **25,403 metric tons of uranium**, essentially all from 1961 to 1999. The IAEA's UDEPO database, which counts the Mounana district from 1962, gives 25,635 tons. An earlier OECD-IAEA study gives 27,872 tons; a 2002 OECD table listing mine-by-mine output sums to 28,492 tons; the mining company Areva's own communications said 26,000 and later 26,660 tons. The differences come from accounting scope — ore at the mine mouth versus delivered product, and whether stockpiles are included. This file uses the Red Book figure of 25,403 tons as the headline and notes the others wherever detail matters.

The ore averaged about 0. At the 1964 Geneva atoms-for-peace conference, Gabon's own delegation presented the district to the world: the Mounana ore, the cold sulfuric-acid leach, the magnesia precipitation, about 120 tons of concentrate a month — the same magnesium-uranate chemistry the isotope analysts would puzzle over in 1972.37 percent uranium, rich for its kind. One small entry in the IAEA record is worth pausing on: of everything COMUF shipped, **94 tons of uranium were isotopically depleted** — the mined remnants of the reactor zones themselves, the physical evidence of the

two-billion-year-old fire, dug out, milled, and sent to France before anyone knew what it was.

Per mine, the best official table (the 2002 OECD study) reads: Mounana open pit 3,915 tons and underground 1,844 tons (1960s–1975); Oklo open pit 7,242 tons (1970–1985) plus underground 8,100 tons (1977–1993); Okélobondo 3,500 tons (1989–1997); Boyindzi 2,691 tons (1980–1991); Mikouloungou 1,200 tons (1997–1999). The district peak was about 1,250 tons of uranium per year in the late 1970s — the uranium boom years, when the world price ran from about six dollars a pound to over forty. Gabon was, for a while, one of the ten largest uranium producers on Earth.

And the product of all this digging went, as Part VII will detail, almost entirely to one country, for both its power stations and — for the first decades — its weapons.

5. Where exactly is this place?

For the record: the Oklo mine lies at roughly 1 degree 24 minutes south latitude, 13 degrees 10 minutes east longitude, about two kilometers south of Mounana village, which is about 25 kilometers north of Moanda, the manganese town, and about 65 to 70 kilometers by road southeast of Franceville, the provincial capital. Bangombé's reactor lies thirty kilometers south of the Oklo-Okélobondo deposits — or, measured the other way, about 25 kilometers west-northwest of Franceville; the literature contains distances of twenty, twenty-five, thirty, and thirty-five kilometers depending on where the tape measure starts, and this file tells you the reference point every time a distance is quoted. The district is not remote by Gabonese standards — the Trans-Gabon Railway reached Franceville in 1986, and the old ore road still runs past the flooded pits — but Bangombé itself, on the land of the manganese plateau, takes permits and a guide, as we will see in Part X.

5. The life that made it all possible

One more resident of these rocks deserves its own paragraph, because without it none of the rest exists. The FB black shales of the Franceville basin carry the remains of an early biosphere — and in 2008, a team led by Abderrazak El Albani described macroscopic fossils from these 2.1-billion-year-old beds: disc- and ellipse-shaped structures up to twelve centimeters across, which they interpreted as organized colonies, possibly among the earliest complex life. The find is debated — some specialists read the same structures as colonies of simpler organisms or even inorganic formations, and the responsible literature says the question remains open — but what is not debated is the broader point: the Franceville basin was biologically busy. Cyanobacterial mats carpeted the tidal flats where the FA sands were laid down; their oxygen dissolved uranium; their buried bodies became the kerogen that trapped it; their descendants' photosynthesis built the atmosphere that made the whole phenomenon possible. When scientists call the algae "the world's first nuclear engineers," it is not a joke. They engineered the ore. The reactors were their downstream consequence, two billion years before the word "engineer" existed.

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PART III — THE PHYSICS: HOW IT WORKED, EXACTLY

This is the heart of the file. Take it in pieces; every step is real, measured, and argued over in the literature, and this file will always tell you where the certainty ends and the debate begins.

1. The window in time

A natural reactor needs its uranium-235 share to be at least about 3 percent. That fixes the whole phenomenon inside a specific window of Earth history. Before about 2.5 billion years ago there was not yet enough oxygen in the environment to dissolve uranium and concentrate it into ore bodies. After about 1.7 billion years ago, uranium-235 had decayed below the critical threshold, and no natural deposit on Earth could ever go critical again no matter how rich or wet it was. Between those two limits — a few hundred million years wide — nature had exactly one chance per suitable deposit. The Franceville ores formed at 2050 million years and went critical at 1950 million years, plus or minus forty — dates fixed by three independent clocks (uranium-lead on the mineralization, rubidium-strontium and argon on the reactor clays, and the fission-product ratios themselves), all agreeing. The famous round number "two billion years old" is exactly right; the variants you will see in popular books — 1.7, 1.8, 2.1 billion — are earlier measurements, popular rounding, or simple error, and the fact-check file tracks them.

2. Reactor anatomy

Each reactor was, by our standards, tiny. The biggest, Reactor Zone 2, was a lens about twelve meters long and eighteen meters across, half a meter to a meter thick — about 240 cubic meters of rock, roughly the volume of a small house. The smallest surviving lenses are a few centimeters thick. Inside the lens sat the core: a layer, five to twenty centimeters thick, of uraninite grains — fifty to eighty percent uranium — enclosed in clay and, in many zones, in bitumen, the solidified residue of the oil that had helped trap the ore.

Around every core sits a three-part clay aureole, like the rind of a cheese: an inner altered clay, a middle magnesium-chlorite shell, and an outer shell of illite (both are clay minerals, and their chemistry records the heat), together about two meters thick. The sandstone around a reactor lost its quartz entirely — dissolved away by the hot fluids — so that the surrounding rock slumped and draped down into the void, a fossil of the volume the reaction consumed. Geologists named this altered jacket the *argile de pile* — the "pile clay," using the French word for a reactor pile. It is the reactor's sarcophagus, and, as we will see, it did its job for two billion years.

3. Going critical

A chain reaction is a simple idea with strict arithmetic. When a uranium-235 nucleus absorbs a neutron, it usually splits, releasing energy and, on average, two or three new neutrons. If at least one of those neutrons goes on to split another nucleus, the reaction sustains; if more than one does, it grows; if fewer, it dies. In pure uranium the neutrons are born too fast, and they escape or are captured without splitting enough; that is why reactors need the neutrons *slowed down* — moderated — by something light, like water, which bounces the neutrons around until they are slow enough to split uranium-235 efficiently.

For the Oklo cores, the arithmetic worked out like this, in the careful computer models built decades later by Salah-Eddine Bentridi and colleagues — the so-called Monte Carlo method, which simulates neutrons one at a time through the rock, millions of times over, and averages the outcomes: with water filling twenty to forty percent of the pore space (the empty gaps in the rock), a core of about five volume-percent uraninite — roughly twenty percent uranium by weight — could reach criticality at a radius of about a meter. No reaction zones are observed below about twenty percent uranium, which is exactly where the models say criticality becomes impossible. Water content mattered critically: the models say the rock needed twelve to fifteen percent water to moderate efficiently — more or less and the reaction cannot start. The poison budget tells the same story from the other side: the reactor-grade sandstones of the district carried lithium around a hundred parts per million,

boron twenty to two hundred and fifty, vanadium two hundred to eight hundred, chromium, nickel and copper five to a hundred, cadmium only a few, rare earths around a hundred, and hafnium under twenty. Every one of those numbers sits below the threshold at which absorption overwhelms a 3.5-percent-fueled core of meter scale — while the neighboring deposits that failed to react exceed the vanadium line tenfold. And the geometry, from the Strasbourg-Batna modeling: an initial core about seventy centimeters thick (derived from a sphere of thirty-five-centimeter radius at twenty percent uranium), porosity twenty to forty percent — the exact range of hydrocarbon reservoirs, which is no coincidence — a bare critical radius of about a meter, cut by roughly a third when the surrounding water-saturated sandstone reflected neutrons back in. Nature's reactor was, dimensionally, a small room of wet rock.

The cores also had to be nearly free of neutron poisons, and as Part II explained, the hydrocarbon trap had filtered the poisons out. Vanadium, boron, rare earths: all at a minimum.

And so, at some moment around 1950 million years ago — probably when a fracture let groundwater into a lens that had just reached the right size — a neutron from spontaneous fission or a cosmic ray started a cascade, and the water in the rock caught the neutrons, and the cascade held. A natural reactor lit off.

4. The pulse: half an hour on, two and a half hours off

Here is the single most wonderful measured fact in this entire file, and it took thirty-two years to extract.

A reactor this small, heating water in rock, could not run steadily. When the chain reaction built power, the water heated; when it heated enough, it boiled; and when it boiled away, the moderator vanished and the reaction stopped dead — a completely self-limiting cycle, the same negative feedback that keeps human-built boiling-water reactors stable, achieved by geology alone. After shutoff the rock cooled, groundwater seeped back in, and two to three hours later the reaction restarted. On, off. On, off. For hundreds of thousands of years.

In 1977, the Soviet physicist Yuri Petrov proposed this self-regulating pulse mechanism, based on the negative void coefficient — the loss of moderation as water turned to steam. But until 2004 there was no direct evidence of the timing. Then Alex Meshik and his colleagues at Washington University in St. Louis did something forensic and beautiful. Xenon is a noble gas — an element so standoffish it almost never bonds with anything — produced abundantly in fission, and it diffuses out of minerals almost immediately on geological timescales — so any xenon still present in the ore must have been trapped the instant it formed, while the reactor ran. Meshik's team found Oklo xenon locked inside tiny grains of an aluminum-phosphate mineral — probably wavellite, a hydrated phosphate that grows only below about 300 degrees — inside the reactor rock, at concentrations up to

three-hundredths of a cubic centimeter per gram, the highest ever measured in any natural material. Aluminum phosphate forms only below about 300 degrees Celsius, which means the mineral grew in place as the reactor cooled, cupping the xenon from each pulse.

And the xenon isotopes told the schedule. Xenon-136 appears within about a minute of fission starting; xenon-134 within about an hour; xenon-132 and 131 over days; xenon-129 only over millions of years. Comparing what was present to what should have been, Meshik's team reconstructed the reactor's duty cycle: **about thirty minutes of operation followed by about two and a half hours of cooling** — a three-hour pulse, hourglass-exact, recorded in a mineral for two billion years. The paper, in Physical Review Letters (volume 93, article 182302, 2004), acknowledged the late Paul Kuroda, with whom the cycling idea had been discussed. Kuroda himself had published his own version of the analysis in 1993, deriving the two-to-three-hour off-time from the xenon data of the 1970s — the credit chain, as this file always insists, is distributed.

Later modeling (a 2020 Imperial College study) reproduced the cycle from fluid physics and noted the honest limitation: the half-hour-on values describe the best-measured material, and different zones, or different ages of one zone, need not have followed exactly the same rhythm.

5. Power, duration, and the fuel budget

The numbers, with their pedigrees:

Power. Each zone ran at an average of ten to fifty kilowatts of heat — the biggest estimates never exceed one hundred kilowatts, which is about the power of twenty-five clothes dryers, or one-tenth of one percent of a modern power reactor. Kuroda's own comparison has never been bettered: the total energy the whole Oklo field produced — about ten to fifteen thousand megawatt-years, by the estimates of the 1975 symposium and later work — is roughly what one large power reactor generates in four or five years, or enough to run a city the size of Budapest for about a year.

Duration. Zone by zone, the reactors ran from twenty thousand years to about eight hundred fifty thousand years. The best-measured: Zone 2, about 620,000 to 850,000 years; Zone 10, about 160,000 years; Zone 9, about 220,000; the Bangombé reactor, on the order of 100,000 to 800,000 depending on the estimate. The popular figure "150,000 years" is a reasonable middle; the widespread claim of "150 million years" is simply wrong, and the best modern popular writing now says so explicitly.

Fuel. Across the field, about five to six tons of uranium-235 were fissioned — which is why two hundred kilograms were missing from the French accounting. George Cowan worked out the fission shares for Scientific American in 1976: about two-thirds of all fissions were of the uranium-235 originally present; most of the rest were of *new* uranium-235 bred from uranium-238 via plutonium-239 — because the reactor bred plutonium in place, and plutonium-239 — whose half-life of 24,000 years means that half of any batch decays away in that

time — decayed back into uranium-235 faster than the slow reaction burned it. A few percent were direct plutonium-239 fissions, and a few percent fast-neutron fissions of uranium-238. The zone-by-zone table, from the Strasbourg group's compilations: fissiogenic plutonium about one to two tons; energy roughly fifteen thousand megawatt-years in the classic accounting; and fluences — the lifetime total of neutrons that passed through each square centimeter of rock, a kind of neutron mileage — up to about 10^21 per square centimeter.

One correction this file must make to the public record: Wikipedia's article on natural reactors carries a per-reactor inventory table claiming about 129 kilograms of technetium-99 and similar masses for other fission products per zone — impossible arithmetic against the four to six kilograms (nine to thirteen pounds) of uranium-235 actually fissioned per zone, and the same page mislabels a whole-field energy figure (100 megatons of TNT, 420 petajoules — correct for the whole field) as the output of a single reactor, which would need fifty tons of fuel. The correct per-zone masses, derived from the standard six-percent fission yields: roughly 250 to 290 grams each of technetium-99, cesium-135, zirconium-93, cesium-137, and strontium-90 per four-and-a-half-kilogram (ten-pound) burn. Both the error and the correction are entered in Appendix B, because this file's rule is that errors are evidence too.

6. What controlled the power

Three regulators worked together, and human reactor designers use versions of all three.

First, the **void coefficient** — the pulse mechanism of Section 4: boil the moderator and the reaction stops. Second, **burnable poisons**: isotopes like samarium-149 and gadolinium-155 that voraciously eat neutrons, building up in the fuel as fission proceeds and slowly throttling the reaction — nature's control rods, which burn away as they absorb, smoothing the power curve exactly as engineered shim poisons do. Third, the **Doppler effect**: as the fuel heats, the uranium-238 resonance absorption broadens, eating more neutrons — the same physics that makes modern reactors inherently stable. The Oklo cores had one more lever: as the reaction consumed uranium-235 and the clay envelope choked the water supply, the reaction simply starved. There is no evidence of any meltdown, explosion, or runaway — the one hypothesis of an ancient "nuclear explosion," entertained by Michèle Neuilly for a few weeks in 1972, was retired by the data. The reactors simmered, pulsed, and quietly died.

7. How hot? An honest debate

You would think the temperature of a fossil fire could be measured. It can, but the measurements argue with each other, and this file presents the argument rather than picking a winner.

The evidence for *hot*: fluid inclusions in quartz record 400 to 500 degrees Celsius; the pile clays carry the high-temperature 2M form of illite; and early analyses, including Damour and Dyson's in 1996 for the physics application we will meet in Part V, assumed 450 degrees or more. The evidence for *cooler*: direct neutron-temperature measurements from lutetium and

gadolinium isotopes give 180 to 400 degrees, with the most careful modern work — Takahiro Hidaka and Philippe Holliger's 1998 study, and the 2014 review by Carl Davis and colleagues — favoring 200 to 300 degrees for the best-preserved zones; and a fresh-core Monte Carlo study by Petrov and colleagues in 2006 found the critical temperature of a brand-new Oklo core at about 725 Kelvin (452 degrees Celsius), above which the water would boil away entirely. Davis's 2014 review, the most authoritative modern summary, states the puzzle plainly: no single temperature fits everything, and the pulsed mode of operation may mean no single temperature ever applied. What is safe to say: the cores ran somewhere between about 150 and 450 degrees Celsius (300 to 840 degrees Fahrenheit), locally higher, with 200 to 350 the best-supported band — hot enough to bake the rock, never hot enough to melt it. One zone, Zone 10, is special: its minerals record unusually *low* temperatures, 100 to 120 degrees, making it a natural cold reactor — and the international programs valued it precisely as the needed second data point.

8. Death and burial

The reactors ended not with a bang but with constipation. Two mechanisms finished them. As the heat and fluids dissolved the quartz (the mineral of sand and glass) out of the surrounding sandstone, the cores compacted — losing up to nine-tenths of their thickness, ending as the thin, rich lenses the miners found — while the newly grown clay envelope sealed the core off from circulating water. And the poisons accumulated. With water shut out and burnable absorbers saturating, the neutron economy closed down, and each reactor froze in its final configuration, mid-story.

Then came two aftershocks, dated to the year by patient isotope work. About 750 to 860 million years ago — the range spans direct mineral dates of 746±16 million (zircon) and 783±55 (samarium-neodymium) on the dyke itself, and a galena-linked 860±39 — a wide wall of molten rock, the dolerite dyke (a dyke is such a wall cutting across the layers; dolerite is a dark volcanic stone), forced itself through the basin, cutting across the old reactors. It drove the first great migration of lead and some fission products: Nagy and colleagues showed that the first major uranium-and-lead-loss episode from the uraninite coincides with that intrusion, 1.1 billion years after criticality. Then, around 650 to 500 million years ago, a Pan-African mountain-building event reheated and recrystallized the rock — the nanocrystalline uraninite textures and the second lead-loss episode date to it. And here is the miracle the waste scientists still cite: through both cataclysms, through two billion years of

groundwater, the great majority of the fission products stayed within meters of where they formed. Uraninite grains kept their actinides — including more than a tonne of bred plutonium — to within a meter or so. The rocks, not any engineer, did that.

9. The zones, one by one — profiles in brief

Because this file promises everything, here is each reactor as an individual, with its own measured biography. The parameters below come from the Strasbourg compilations and the per-zone studies cited in Appendix C; where two measurements disagree, both are shown.

Reactor Zone 2 — the big one. The largest and best-studied core: a lens about twelve by eighteen meters, twenty to fifty centimeters thick, some 240 cubic meters. About 1,800 kilograms of uranium-235 — about two tons, a car's weight in uranium — were fissioned here over some 600,000 to 850,000 years — a large share of the whole field's burn. Its fluence, about 9×10^20 neutrons per square centimeter, and its fission shares (about 95 percent from original uranium-235, 2 percent from uranium-238 fast fissions, 3 percent from plutonium) are the reference values the textbooks quote. In 1975 the world's scientists sat on benches at its edge; in 1976 a section of it was pinned to the pit wall so mining would not erase it; today that pinned face and a wall of archived cores are all that remain of it.

Reactor Zone 9 — the puzzle that started a school. Found at the end of 1978, three hundred meters south of the northern group: a small lens, seven by twelve meters, dipping at 45 degrees, only a few centimeters thick today — and, by the

modeling of Bentridi and colleagues, originally perhaps ten times thicker before the rock compacted. Its modest size and closed, organic-rich environment made its ignition hard to explain, and that honest puzzle is what drove the modern Monte Carlo work that finally reproduced small-reactor criticality. Its measured fluence is about 3.6×10^20, its fission shares roughly 88/8/4 percent, its duration around 220,000 years.

Reactor Zone 10 — the cold reactor. Discovered in 1982, deep at about 400 meters, Zone 10 is the field's outlier: its mineral assemblage records operating temperatures of only 100 to 120 degrees Celsius, against the 300 to 450 of its siblings. About 650 kilograms (1,400 pounds) of uranium-235 burned here over some 160,000 years. The international waste programs prized it exactly for its coldness — a second data point, a reactor that ran under conditions closer to what a repository designer would choose.

Reactor Zone 13 — the extreme one. Found in 1984, sitting ten to thirty meters from the great dolerite dyke that later disturbed it, Zone 13 holds the records: the highest fluence (about 1.2×10^21), the shortest inferred duration, the most fast-neutron fissions (about 18 percent of the total — direct evidence that at least one Oklo core ran with a noticeably harder neutron spectrum), and, measured in 2018 by the Naval Research Laboratory, the most depleted natural uranium known anywhere: 0.3655 percent uranium-235, barely half the normal value. Its cesium had been flushed within about three years of shutdown; its technetium stayed locked in metal. Zone 13 is where the Oklo story's extremes live.

Reactor Zone 16 — the apatite vault. Sampled from the pit wall and by drilling, Zone 16 is the one where fluoroapatite — a common phosphate mineral — was caught in the act of sequestering plutonium and the rare earths, with one apatite fraction even showing slightly *enriched* uranium (a fingerprint of plutonium having been captured and then partly decayed). The 1996 Cadarache study of this zone remains a standard citation for phosphate-based waste forms.

OK84, the Okélobondo deep reactor. A kilometer and a half south of the pit, at 300 to 450 meters depth — deep enough to be the field's best stand-in for a real repository, which is why the Swedish program concentrated its boreholes there. Less weathered than anything at the surface, rich in organic matter, it still holds its story nearly intact.

Bangombé — the survivor. Twelve meters down, a lens a few centimeters thick, perhaps ten meters across, its core still resting in its clay envelope under the protection of a 2017 Gabonese decree and a UNESCO tentative listing. It burned for somewhere between one hundred thousand and eight hundred thousand years, at modest power, in sandstone stained dark with manganese. Of the sixteen, it alone was never mined — too shallow to be worth the digging — and so it alone remains, the last natural reactor on Earth, waiting for the next permitted scientist.

And of the numbered candidates that were never confirmed — zones 11, 12, and 14 — this file records them too, as the honest ledger of a search that numbered everything and verified most: sixteen candidates, fifteen confirmed at Oklo-Okélobondo, one at

Bangombé, and sixteen reactors in all.

When the last reaction died, the only life on Earth was single-celled. The planet's first nuclear engineers had been the algae whose oxygen made the ore possible — the writer James Lovelock put it best: an unassertive community of modest bacteria built a set of reactors that ran for hundreds of thousands of years. The reactors then rested, intact, through the rise of oxygen, the snowball Earth (periods when ice may have covered nearly the whole planet), the Cambrian explosion (the sudden flowering of complex animal fossils about 540 million years ago), the dinosaurs, the mammals — until a French chemist noticed three-thousandths of a percent missing.

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PART IV — THE EVIDENCE, ISOTOPE BY ISOTOPE

How do you prove a rock used to be a reactor? Not by looking at it — the ore looks like ore. You prove it by weighing the atoms, one isotope at a time, and showing that their proportions are impossible by any natural recipe except fission. This part walks through the fingerprints in the order they convinced the world, with the people and the machines attached.

1. The uranium itself

The opening fingerprint was uranium-235 depleted from 0.72 percent down to 0.44 percent, and in the most extreme sample ever found — from Reactor Zone 13, measured in 2018 by a Naval Research Laboratory team — down to 0.3655 percent by weight. Individual ratios vary wildly: one famous sample, known as Oklo M, measured 0.44 percent; the Dozol archive records yellowcake batches at 0.625 and 0.441 percent and ore samples at 0.592 and 0.625. The depletion pattern, concentrated in the richest ore and fading to normal within a couple of meters of each zone's edge, is exactly what neutron physics predicts: a reactor burns its fissile atoms in the center and fades at the margins.

The other uranium isotopes behaved as they should: uranium-234 showed no anomaly (it was in equilibrium with its parent), and uranium-236 was expected from fission but has long since decayed away — its absence, as Part I noted, is what cleared the theft-and-recycling hypothesis.

2. Neodymium: the first confession

Neodymium is a rare earth — one of the chemically similar family whose names sit together at the bottom of the periodic table — and fission makes it in a fixed isotopic recipe that looks nothing like natural neodymium. Natural neodymium is 27 percent neodymium-142; fission makes almost none. So when Guy Nief's team at Saclay separated the neodymium from an Oklo ore sample in July 1972 and found almost no neodymium-142 — about 1.4 percent instead of 27 — with the other isotopes jumbled into exactly the fission pattern, the case closed. Dozol's team at Cadarache ran roughly a hundred such analyses on their spark-source mass spectrometer, developing rhenium-filament chemistry that preferred neodymium over samarium, so the fingerprint could be read cleanly even in tiny samples. The neodymium-143 and neodymium-145 excesses, corrected for neutron capture, matched the thermal-neutron fission of uranium-235 — thermal meaning the slowed-down kind that the water made possible — quantitatively — not approximately, but to within the measurement error.

For readers who like the numbers themselves, here is the original 1972 fingerprint (atom percent), from the Neuilly note as tabulated in the modern reviews — natural neodymium, the fission recipe, and the Oklo sample side by side:

Isotope	Natural	From U-235 fission	Oklo sample
142Nd	27.1	~0	1.4
143Nd	12.2	28.8	22.1
144Nd	23.9	26.5	32.0
145Nd	8.3	18.9	17.7

Isotope	Natural	From U-235 fission	Oklo sample
146Nd	17.2	14.4	15.6
148Nd	5.7	8.3	8.0
150Nd	5.6	3.1	3.4

Look at the first row: the isotope fission never makes is the one that all but vanished. That single row, measured in a French laboratory in July 1972, is the discovery — everything since has been the elaboration.

3. Samarium-149 and the shadow of a constant

Samarium-149 is the strongest natural neutron absorber, and a reactor burns it away preferentially — so a depleted samarium-149 is a reactor's signature. The Oklo samples were extraordinary: ratios of samarium-149 to samarium-147 as low as about 0.02, against a natural value near 0.9. It was this very fact — the precise amount of samarium-149 left behind depends exquisitely on the exact energy of one nuclear resonance — that let a young Soviet theorist named Alexander Shlyakhter propose, in a 1976 letter to Nature, a use for Oklo that nobody had imagined: testing whether the constants of physics themselves have changed over two billion years. That story gets its own part, next.

4. Ruthenium and technetium: the ten-million-year delay

Ruthenium in the Oklo cores is measurably strange: natural ruthenium is 12.7 percent ruthenium-99, but the reactor ruthenium runs 27 to 30 percent. The excess is the fossil of technetium-99 — a fission product with a 210,000-year half-life that decayed into ruthenium-99 long after the reactors died. Here is the elegant clock inside that fact: by measuring how much ruthenium-99 had accumulated from the technetium, researchers could time the *migration* of fission products — the element-fractionation happened within about a million years of shutdown — and by measuring the tiny direct ruthenium-99 fraction, they could tell how long the reaction ran, since technetium builds up only while fission continues. The Los Alamos team of the late 1970s — Apt, Bryant, Cowan, Daniels, Vidale, with Douglas Brookins of the University of New Mexico — searched a dozen other Precambrian uranium districts on four continents for the same fingerprints, from Shinkolobwe in Zaire to the Northern Territory of Australia to Saskatchewan. They found nothing. Oklo remains the only one ever found — a point that will matter in Part IX.

5. The xenon hourglass

You met the xenon in Part III — the pulsing clock in aluminum phosphate. To the retention story it adds this: the xenon and krypton were *not* retained in the uranium mineral; they escaped to the last place available, the phosphate that grew as the reactor cooled, and there they sat for two billion years. Chemically fractionated fission xenon — xenon whose isotope recipe was scrambled by the migration of its tellurium and iodine parents — was recognized in Oklo samples by the Soviet school in the 1970s and studied in depth by Meshik; the same signature appears in meteorites, in ocean-floor glasses, and in ancient Greenland rocks, which is how a two-billion-year-old reactor ended up contributing to the science of the early solar system.

6. The metallic fossils

Under the microscope, the reactor ores hide flecks of metal — aggregates, some as large as a tenth of a millimeter, of ruthenium, rhodium, palladium, technetium's descendants, molybdenum, and the platinum-group metals: the so-called epsilon phase, the same alloy family that forms in irradiated nuclear fuel. In 2018, the Naval Research Laboratory team led by Evan Groopman imaged these aggregates isotope by isotope and found they had captured cesium and barium at the moment of the reactor's death — locked in metal within about five years of shutdown, sealed against everything since. And the mineralogy itself keeps the radiation-damage record: Marina Deditius and colleagues showed that phosphate-bearing coffinite — a uranium silicate mineral — from Bangombé carries the amorphization scars of two billion years of self-irradiation — a materials durability test no laboratory can run. In 2021 the same group found that nearly all the technetium was retained in these metal alloys, while nearly all the cesium had been flushed out of the fuel within about three years, save for what the metals trapped. These tiny metal fossils are now the reference standard for how spent fuel's most troublesome elements might behave over geological time.

7. What left, and how far

The honest map of retention, element by element, from three decades of isotope work — the Brookins analyses, the Hidaka-Holliger 1998 survey, the Groopman work, and the Canadian and American government reviews:

Kept in place, essentially fully: uranium, neptunium, plutonium, americium, thorium; the rare earths; yttrium; niobium; zirconium (mostly — but see below); ruthenium, rhodium, palladium, tellurium (in the metal aggregates); technetium (in the aggregates and in surrounding sulfides, sulfur-bearing minerals).

Kept locally, within meters: lead (about half retained); the alkalis and alkaline earths — rubidium, strontium, cesium, barium — which escaped the uranium mineral but were fixed by the pile clays, by apatite and other phosphates, and by secondary carbonates within one to two meters.

Lost to the wider rock: xenon and krypton (except the phosphate-trapped fraction); iodine; cadmium, nearly entirely; silver and tin, more than half; molybdenum, heavily under oxidizing conditions.

And the exception that proves the rule: zirconium — long treated as the immobile standard of geochemistry — was found by Rodolphe Bros and colleagues in 2003 to have migrated through the alteration halo over meter distances, carried in hot fluids while the reactors ran. Oklo broke even that assumption.

The quantitative summary used by the American regulatory literature: for the retained elements, fractional loss rates of about 10⁻^9 per year; for the mobile oxyanions, 10⁻^7 to 10⁻^8 per year locally; for lead and the uranium mineral itself, about 10⁻^10 per year. Two billion years of groundwater — and the actinides, the elements regulators worry about most, never left the building.

8. The analysts, named

8. A footnote on the periodic table

One consequence of the fingerprints reached further than anyone expected. Because Oklo ores contain elements whose isotopic compositions lie outside anything natural uranium can make, the international chemistry union's atomic-weight tables carry, to this day, a special mark — footnote "g" — on twenty elements, from antimony to zirconium: a note that geological materials exist in which the element's isotopic recipe falls outside the normal range. The Oklo reactors changed the fine print of the periodic table. When the same isotopic work needed a cross-section convention to express its neutron measurements, the standard it cited — the Westcott formalism — was itself a Chalk River report from 1958; the physics, like the ore, kept finding older foundations.

Because this file names names: the isotopic evidence was built by Michèle Neuilly, Jean-François Dozol, and Jean-Claude Nimal at Cadarache; Guy Nief, Georges Vendryes, Jacques Yvon, and Jean Bussac at Saclay; Etienne Roth's team; Wilhelm Maeck, John Delmore, and the Los Alamos migration group under George Cowan, Don Curtis, and Alex Gancarz; the Strasbourg school — Gauthier-Lafaye, Weber, and later Benoît Gall; Takahiro Hidaka's Nagoya group across three decades; Alex Meshik, Charles Hohenberg, and Olga Pravdivtseva at Washington University; Evan Groopman at the Naval Research Laboratory; and the Soviet and Russian school — Yuri Petrov, Alexander Dymkov, and the late Alexander Shlyakhter. The full

citation record, several hundred entries, is Appendix C.

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PART V — THE CONSTANT: WHAT OKLO SAYS ABOUT THE LAWS OF PHYSICS

In 1976, Alexander Shlyakhter, a young theorist at the Leningrad Nuclear Physics Institute, noticed something almost arrogant in its simplicity. The samarium-149 nucleus captures slow neutrons through a resonance — a special energy where capture is vastly enhanced — sitting at 97.3 milli-electron-volts today. Exactly how much samarium-149 a reactor burns therefore depends on the exact position of that resonance, and the resonance's position depends, in part, on the fine-structure constant — the number that sets the strength of the electromagnetic force, one of the deepest constants in physics. If that constant had been even slightly different two billion years ago, Oklo's samarium would carry a different fingerprint. Measure the samarium, and you measure the constant — at two billion years' remove. His 1976 letter to Nature proposed exactly this, and gave the first bound: the fine-structure constant could not have differed by more than about one part in ten million.

The method was refined over thirty years, and this file presents the whole arc, including its one dramatic detour. Thibault Damour and Freeman Dyson calculated the resonance sensitivity properly in 1996 and bounded the change at roughly ±10⁻^7. Yasunori Fujii and colleagues in 2000, and then Steve Lamoreaux and Justin Torgerson in 2004, re-did the neutron-spectrum treatment — and Lamoreaux, using a more realistic neutron spectrum than the standard bell curve (what physicists call the Maxwellian) that earlier analyses had

assumed, announced a *detection*: the constant seemed to have been smaller back then by about 45 parts per billion, a six-sigma claim that made international news under headlines about the speed of light possibly changing. It was, and remains, a lovely illustration of how science corrects itself: three years later Lamoreaux's own group, with Charles Gould and others, reanalyzed with better samarium data and retracted the detection — the bound collapsed back to roughly ±10⁻^7, consistent with no change at all. The Russian school — Petrov, and later Onegin and colleagues — pushed the modeling further still, down to about 10⁻^17 per year, and added an independent thermometer from lutetium isotopes. The 2014 review by Davis, Gould, and Sharapov remains the standard summary: within its uncertainties, nuclear physics two billion years ago was the same as today; the fine-structure constant is constant to about a part in a hundred million over that span; and the temperature of the reactors, it turned out, is the largest remaining uncertainty in the whole calculation — which is why Part III presented that debate in full.

What does this mean, for a reader who is not a physicist? The laws of physics are written into the Oklo rock with a precision no laboratory on Earth can reach for that timescale. And there is a pleasing footnote: the same isotope-resonance technique has been proposed for hunting physics beyond the standard model — tests of time-reversal symmetry and of quark masses — using Oklo as the laboratory. Nature's reactor turned out to be an instrument. And there is a second, stranger way Oklo's ghost still haunts physics. When the KamLAND detector in Japan and

Borexino in Italy began counting *geoneutrinos* — the antineutrinos, ghostly particles that radioactivity emits and that stream through almost everything, including the whole Earth — they were measuring, indirectly, the planet's hidden engines. The mainstream reading of their data puts tight limits on any natural reactor at the Earth's core: a few terawatts (trillion-watt units) at most, where the fringe "georeactor" hypothesis wants three to five. The hypothesis's own simulations concede the constraints, and its critics note it has no confirming evidence. But the whole debate exists because of Oklo — because once you have seen a natural reactor in a rock, you are entitled to ask whether the planet ever ran a bigger one, and the neutrino detectors are the instrument that answers. Today the answer is: probably not, and if so, only faintly.

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PART VI — THE WASTE LESSON

By the second international meeting in 1977, the framing had shifted, and it shifted for a practical reason: the world's nuclear programs were waking up to the problem of what to do with spent fuel that stays dangerous for a hundred thousand years. If you wanted evidence that waste could stay put in rock for geological time, here was a two-billion-year experiment, already run. The Oklo Natural Analogue programs — first under the European Commission with French leadership from 1989, then a second phase in the 1990s, with Canada, Sweden, Switzerland, Britain, Japan, and the United States joining through the IAEA's working groups — spent a decade and substantial funding measuring exactly that question. Oklo stood at the head of a whole family of natural analog studies — Cigar Lake in Canada, Alligator Rivers in Australia, Poços de Caldas in Brazil, Palmottu in Finland — the programs from which the field learned its methods.

What the programs found, in the balanced form this file owes you:

The case *for* the analogy: the great majority of actinides and many fission products genuinely did remain localized for two billion years; the pile clays, the phosphates, the metal alloys, and the bitumen each demonstrably trapped specific elements; the geological setting — a stable craton (an old, tectonically quiet block of continent), impermeable clays, permanently oxygen-free chemistry — proved its worth through two thermal cataclysms and eons of groundwater. The quantitative loss rates cited in

Part IV are, for the retained elements, astonishingly small.

The case *against* pushing the analogy too far, stated by the programs' own reviewers: Oklo is a *site*, not a *design*. The Canadian regulator's review put it in writing in 1986 — the database, vast as it is, gives a site-specific description, and generalizing it requires reconstructing coupled heat-fluid-chemistry processes no one can yet model quantitatively. The American waste program's own documents ask the sharpest questions: although "most" fission products were retained, what precisely is that fraction, over what geometric scale did the lost percentage disperse, and could the major chemical alteration that hit some of the Oklo rocks about ten million years ago have been predicted even a million years in advance? A French-mirrored American document delivers the verdict this file finds fairest: Oklo provides *a good presumption, but not a demonstration*. The Swedish program that drilled Bangombé said the quiet part out loud in its own annual report: to date, direct use of analogue data in performance assessment had not been undertaken — the linkage, not the data, was the missing engineering. And the critics' oldest point still stands, from a 1983 symposium: Oklo had no mine shafts to seal, no engineered barriers, no human error — and some of its zones would never be licensed as a repository today.

The bacteria in the file

One line of evidence deserves its own paragraph, because it surprises everyone who meets it. When the Swedish nuclear waste program drilled its boreholes at Bangombé in 1993, 1994, 1996, and 1998, the cores came up not just wet but *alive* — on the order of a hundred million to ten billion bacteria per liter of groundwater, iron-reducers and sulfate-reducers eating the organic matter in the shale, consuming oxygen, and raising the water's alkalinity. The microbes matter to the waste question for a simple reason: a repository's chemistry is partly run by its inhabitants. Bacteria that strip oxygen out of seeping groundwater help keep uranium in its insoluble, immobile form; bacteria that generate alkalinity help minerals precipitate and trap migrating atoms. The Oklo system, it turns out, did not seal its waste with rock alone — the food web that helped build the ore is still, two billion years later, helping to guard it. Scientists found the iron-oxidizer *Gallionella* growing on the flooded mine walls at Okélobondo. Life engineered the ore, ran the reactors' preconditions, and still tends the grave.

Both cases are evidence. The waste programs still cite Oklo — Finland, Sweden, France, Canada, Switzerland, and the American repositories at WIPP and Yucca Mountain all built analog studies on it — and they cite it with the caveats attached, which is exactly how evidence should be used.

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PART VII — THE MONEY

Science does not happen in a vacuum; it happens in a world of contracts, weapons programs, railway loans, and presidents. This part follows the money, plainly, with the documents named.

1. Who owned the ore

COMUF was chartered in February 1958 as a Gabonese company with French capital — the CEA and a consortium of nine French mining houses. At independence in 1960, the arrangement continued: France was the sole customer, and the 1960 cooperation agreements, renewed over the decades, classified uranium among the "strategic raw materials" that Gabon committed to supply France on a priority basis. In 1974, President Bongo's government took a 25.8 percent stake, and the French state mining group Cogema (created 1976, later folded into Areva and then Orano) took majority control in 1982 with 68.4 percent; a small remainder sat with private holders. Areva became the operator in 1986. The ownership barely mattered to the flow of ore: Gabon's uranium went to France — about 80 percent of exports — with small amounts later to Italy and Japan. By the mid-1980s Niger and Gabon together supplied roughly a third of all French uranium imports, and Gabon's mines were majority-French-owned at every stage.

2. The weapons link

Here the file must be precise, because this is where memory and evidence part company. The enrichment plant where the discovery happened, Pierrelatte, was France's *military* enrichment plant — built by decision in 1958, operational from 1967, dedicated to producing highly enriched uranium for the French deterrent until 1996, when France stopped producing fissile material for weapons. Its uranium hexafluoride feed came from the companion Comurhex conversion plants. Gabonese uranium was routinely screened, milled, and enriched within a complex whose primary purpose was warheads; French military strategists of the era called Gabon "indispensable" to building the force de frappe, and Gabonese uranium certainly flowed into the same enrichment cascades that produced weapons material. What no public record establishes is a tonnage-by-tonnage split between Gabonese ore that ended in weapons and Gabonese ore that ended in power stations — the cascades did not keep such open books. This file therefore records the link as documented — an integrated military-industrial chain, with Gabon as its feedstock anchor — and marks the precise tonnage split as unknown in the open record.

3. 1974: "Take it or leave it"

In February 1974, President Omar Bongo arrived in Paris to renegotiate the cooperation accords and did something no African head of state had done: he announced, at a press conference, that Gabon was raising the price of its uranium — effective immediately, take it or leave it. The French government took it. Nine new cooperation agreements were signed within days. The timing was everything: the 1973 oil shock had detonated the uranium market, the price graph climbing from about six dollars a pound in 1972 to a plateau above forty by the late 1970s, and Gabon's production peaked in exactly those years. Behind the scenes, the files of the French presidency show uranium at the center of Bongo's visits in 1974 and 1975, and his government had quietly studied selling to the United States or Iran as leverage. An American government report of 1976 noted the move and even rumors of Gabon stockpiling unsold production to wait out the market. The uranium — and the manganese, and the oil — gave a small country a lever, and Bongo pulled it.

The same uranium politics ran darker elsewhere in the Francophonie: in Niger, President Diori pushed a similar price line and was overthrown by a coup in April 1974, which France declined to stop. Gabon's contrast case is instructive and is recorded as evidence, not accusation: France had paratroopers in Gabon within a day of its own 1964 coup attempt, restoring President Mba — the difference, the strategic files suggest, being what sat under the rainforest.

4. The railway

The boom paid for the Trans-Gabon Railway — 670 kilometers of standard-gauge line from the port of Owendo through the rainforest to Franceville, started in 1974 with the oil windfall, finished in stages through 1986, at a cost of about 1.6 billion dollars, with French, West German, and later British money. Its rationale was manganese from Moanda, but the uranium rode it too, and the line ran, by no coincidence at all, straight through Bongo's home province to his home town. Scholars of the period call the route choice patrimonial; the trains still run, and the old ore road from Mounana to the railhead is still there.

5. What the company said, and what it cost

At its 1988 peak, COMUF employed over 1,400 people and told the Uranium Institute in London that its thirty years had delivered nearly 20,000 tons of uranium, with two decades of reserves ahead. What the future actually held was the price collapse of the 1980s — forty dollars a pound down to twenty — which squeezed the high-cost deposits and ended with the June 1999 closure, "for lack of economically exploitable reserves," as the official phrasing put it. The cleanup, when it came, was funded heavily by the European Union — more than 50 million euros to Gabon's mining sector since 1997, of which 7 million euros went to the Mounana rehabilitation, completed in 2004 — with the mill dismantled, the tailings dammed and covered, and a monitoring program left behind. Today Orano's Gabonese successor subsidiary still exists, now devoted to care and exploration rather than mining, and the Red Book records the country's remaining resources at about 4,800 tons of uranium, uneconomic at present prices. The mines are flooded. The ore, for now, stays in the ground.

And one quiet thread ties this part back to Part I: the price spike Bongo rode began the year after the discovery, in a market that would never have noticed two hundred missing kilograms if the 1972 price had not made every gram accountable. The bomb-material panic that launched the investigation was, in the end, a bookkeeping anomaly of a boom.

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PART VIII — THE PEOPLE

A reactor file that only counts atoms would be dishonest, because between the atoms and the laboratories stood human beings: the geologists who figured it out, the miners who dug it, and the men and women who breathed its dust for forty years. This part gives the investigators their due and then turns, carefully and without blame, to the hardest chapter of this file.

1. The investigators

The Franceville Project, which ran from 1973 and coordinated the international science, was led on the French side by Roger Naudet, a CEA physicist whose 1991 monograph *Oklo: des réacteurs nucléaires fossiles* remains the standard physics reference. The geology belonged above all to François Gauthier-Lafaye and Francis Weber of the University of Strasbourg — Gauthier-Lafaye the chief geologist of the project, whose 1986 thesis is the definitive account of the deposits, and who, half a century after the discovery, opened the fiftieth-anniversary symposium in Strasbourg in 2022 with the words: sixteen reaction zones, 1.95 billion years, and questions still unanswered. The isotope work was Neuilly's and Dozol's and Nief's; the migration studies were Cowan's Los Alamos group, Brookins at New Mexico, and the Strasbourg cadets — Rodolphe Bros and his colleagues at Cadarache; the Japanese line was Takahiro Hidaka of Nagoya University, whose three decades of Oklo papers trained a generation; the xenon detective was Alex Meshik; the fresh eyes of the 2010s belonged to Evan Groopman at the Naval Research Laboratory, to the Algerian school of Salah-Eddine Bentridi, and to Benoît Gall, who now leads the Strasbourg group; and the theoretical frame that made it all make sense was Paul Kuroda's. The Gabonese were there too — researchers like Evaldie Durastanti Rabnga Mombo working within the Strasbourg team today, and the mine's own chief geologist in 1975, J. P. Bindeman, lecturing to the world from the pit edge. Science, like mining, is people.

2. The miners: what the record shows

COMUF ran the Mounana mill from 1961 and the mines until June 1999. Across those thirty-eight years, more than twenty-five thousand tons of uranium left the district. The question this part must now face is what the work did to the people who did it. The evidence comes in layers, and this file presents each layer in turn, exactly as it stands, with the names and dates attached.

The protection record, from the company's own archives. The earliest instruction in the COMUF files is dated May 5, 1961 — safety orders on radiation protection, approved by the Gabonese mining director the following month. Over the following decades the files record radon measurements, film-badge programs, and periodic reports from the mine's radiation officer, Christian Guizol. The historian Gabrielle Hecht, granted access to the COMUF archives in 1998, found them voluminous but disordered; she could not establish whether Gabonese state inspections ever occurred, and she says so plainly. Several features of the record are, however, documented beyond dispute: workers wore personal dosimeters, but the results were not systematically communicated to them — a 2007 survey of 481 former workers by Sherpa, a French public-interest law group, found 455 of them reported receiving no information about radiation or radon risks at all; the mine's own physicians did not receive Western laboratory results because the protection service did not transmit them, according to Dr. Angélique Kombila, a company doctor until 2004; and the

Gabonese physician Steeve Ondo, who served on the health observatory's scientific committee, explained the statistical heart of the matter on the record: the French occupational-disease table used for screening listed only thirteen radiation-linked pathologies, so a miner could be genuinely sick from his work and still fail every legal test for recognition. A CEA study of Mounana's radon levels in 1984-85 (by Nicole Fourcade and M. C. Robe of the institute's mine-protection laboratory) measured 2.5 to 5.8 picocuries per liter (the picocurie is an older unit, about one decay every three seconds per liter) — averages within the range of French comparison sites, a point the company's defenders raised, and one this file records alongside everything else, because it is evidence too.

The sickness, as the workers report it. By the union collectives' count at the end of 2016, 367 former workers of MATRAC — the association of Gabonese ex-miners, with 1,618 members — had died, many of respiratory disease. A 2015 internal Areva health-directorate email, verified by Agence France-Presse, acknowledged "many serious diseases among former employees, for example contagious tuberculosis." Surveys by Médecins du Monde found more than half of interviewed Gabonese workers reporting pulmonary problems. What no one has produced is a completed epidemiological study establishing cause: the company notes that its own screening of 650 former workers found no declared occupational disease; the workers note they stopped attending screening because results were never given to them; and the anonymous verdict of a

former senior executive, quoted by AFP in 2017, stands as the honest summary of the impasse: that there was radioactivity at Mounana is a reality — but to what degree and to what extent the workers were affected will be very difficult to establish.

The tailings and the river. The physical contamination is documented by the company's own measurements and by the CRIIRAD laboratory (the independent French radiation-monitoring group): through 1975, about two million tons of radioactive tailings — the crushed waste rock left after the valuable ore is extracted — were discharged directly into the Ngamaboungou creek and the Mitembe river; four million more tons went into the Mounana open pit. A 2002 COMUF study, presented to the IAEA, measured soluble radium-226 in the Mitemba at 3.2 becquerels per liter (a becquerel is simply one atomic decay per second) — nearly ten times the international reference — and soluble uranium-238 at 1.7 milligrams per liter, over 180 times the reference. A 2009 CRIIRAD investigation found radiation levels two to fifty times background at public places — a maternity ward, a school, the market built on tailings sand — and the story of the market is worth telling exactly as it happened: COMUF donated a concrete slab to the town; the first pour used sand from the mine area; a radiation check found the slab hot; the company poured a second layer over it. The mayor's comment is on record: the floor is covered — but the walls are not. In 2009 roughly 200 contaminated houses were demolished under a joint program. The company's position, stated publicly many times, is that remediation met international standards, verified by four IAEA missions between 2001 and

2006; the CRIIRAD position is that the reclamation was inadequate. Both are entered here, in full.

The legal road. In January 2016, the MATRAC movement was received at the Gabonese prime minister's office. On January 30, 2019, MATRAC sued COMUF before the civil court in Libreville, alleging exposure to chemicals and radiation; the case was dropped on a procedural irregularity in May 2019, refilled, and on November 17, 2021, the court dismissed the association's claim. Orano's own financial filings state the company's position in its own words: the evidence disclosed to date does not demonstrate either damage attributable to COMUF or the admissibility of such a claim — while adding, carefully, that given the partial nature of the evidence, Orano cannot exclude revisiting its position if new elements emerge. In June 2022, a new collective, Mounana Ura 235, filed a fresh action through the Libreville firm Bourbon et Associés, denouncing decades of neglect; its outcome is not yet in the public record. In France, two families of deceased French expatriate workers were compensated in 2011 — including Jacqueline Gaudet, the association Mounana's founder, who lost her father, her mother, and her husband to cancer after the family lived in a house built from mine tailings; she brought two amendment bills to the French National Assembly, both unsuccessful.

The framework problem. Underneath every one of these struggles sits a technicality with a human face: compensation rides on official lists of recognized occupational diseases — thirteen pathologies in the French table of 1984, a shorter list in

the Gabonese CNSS (the national social-security fund) — while the medical literature counts more than twenty radiation-linked cancers. The result, documented by scholars including Nestor Engone Elloué, who analyzed Mounana as a case of "three-dimensional injustice" — economic, social, and judicial — is that claims fail not because they are disproven but because they do not fit the categories. The American comparison puts the stakes in perspective: under the Radiation Exposure Compensation Act (RECA), by March 2025 the United States had paid 943 million dollars to 9,445 former uranium workers and their families, at 100,000 dollars each. The Gabonese workers' 1,618 complaints remain, as of the latest record, without a single payment.

What this file does not do. It does not assign blame. It notes that the company built a monitoring program in 1996 whose board then declined to publish the results, for fear of "biased analysis and polemics," in its chairman's words; that workers wore dosimeters whose readings they never saw; that physicians were not trained in uranium medicine; that the tailings went into the river for fourteen years; and that the people who did this work were, in the words of one of them, working blind, without ever being told the danger. It notes equally the company's documented counterpoints: the dosimetry program existed, the radon averages matched French ranges, the screening found no declared cases, and the causation chain is genuinely hard to prove forty years later. Both sets of facts are real. The reader is trusted with them. A 2026 paper in Frontiers in Public Health by a Georgetown University group

placed Mounana alongside Chernobyl and Fukushima in a comparison table of radionuclide disasters — unrated on the international scale, containment listed as "none," health impacts recorded as high rates of respiratory illness and suspected cancers with a lack of clinical tracking. Whether that framing will withstand scrutiny time will tell; this file records it as the strongest peer-reviewed statement to date, alongside the company record, the court record, and the workers' testimony, and leaves the verdict history is still writing.

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PART IX — THE CLAIMS

Now we come to the part of this file that some of you opened it for, and this file's promise requires handling it with care: we document every claim as evidence — including the ones the evidence contradicts — because a claim, true or false, tells you something true about the people making it and the age that produced them.

1. The question, stated fairly

Could ancient people have found, understood, or used a natural nuclear reactor? It is not a stupid question, and this file will not pretend it is. It has three parts, which this part answers in turn: *when could a natural reactor have existed* (physics), *what could people have done with one* (engineering), and *what do the claims actually say and where do they come from* (evidence).

2. The physics of the question

A natural reactor requires uranium-235 at three percent or better. That fixed a window in Earth's history — roughly 2.5 billion years ago, when oxygen first made uranium mobile, to 1.7 billion years ago, when decay dropped the share below critical. The Oklo reactors ran 1.95 billion years ago, inside that window. The earliest anatomically modern humans are about three hundred thousand years old; farming is about twelve thousand years old; writing, five thousand. The reactor window closed **one point seven billion years before any possible person existed**. Every clock that dates Oklo — uranium-lead, rubidium-strontium, argon-argon, the fission products themselves, plus the Oklo samarium measurement that bounds changes in the constants of physics, presented in Part V — agrees on this. The only way to put a reactor in human time is to discard all of that dating, and this file examines even that move, in section 8 below.

There is one more physical fact worth weighing: even if timing had allowed it, the Oklo reactors sat two to four kilometers underground when they ran. Nothing on the surface would have shown except, perhaps, warm springs. The only visible sign in 1972 was a yellow crust of oxidized uranium at the pit floor — and it took a precision isotope analysis to know it meant anything.

3. The engineering of the question: who, and for what

Indulge the counterfactual honestly, because it is instructive. Suppose a shallow reactor — something like Bangombé's twelve-meter depth — had existed within human history. What then?

Who could have reached it? Any Iron-Age state with organized mining. The ancient silver mines at Laurion in Greece reached 119 meters by 400 BC; Roman workings in Spain ran comparable depths. Twelve meters is nothing. So: any Bronze- or Iron-Age mining society — Laurion's Athens, Rio Tinto's Rome, the later kingdoms of Africa — could have dug to it.

What would they have found? A warm spring, if groundwater crossed the heat; yellow and green uranium minerals usable as pigments; perhaps an odd sense of the place. At twenty kilowatts — one zone's output — a spring flowing a liter a second would warm by about five degrees: a genuinely warm, noticeable, sacred-feeling spring.

What could they have used? That is where the counterfactual stops. Twenty kilowatts is one household water heater, delivered through solid rock with no pipes capable of bringing pressurized steam to the surface — the aeolipile of Hero of Alexandria, the most sophisticated heat engine of antiquity, was a spinning curiosity that could not be geared to do work, as every historian of technology agrees. Metallurgy, medicine, chemistry: none could draw on the heat. And the exposure: standing at an exposed ore face at Bangombé depth, a worker would absorb,

over weeks, doses that modern medicine associates with hair loss, nausea, and skin burns — the very symptom list, this file notes with care, that appears in ancient texts the fringe literature quotes as eyewitness accounts of ancient radiation. The same chronic-exposure regime, at far lower dose, makes today's residents of Ramsar, Iran — where natural background radiation runs far higher than Mounana's — a living study population for exactly the question such a warm-spring people would have faced. The symptoms are real radiation symptoms. But an ancient people could no more have *used* the reactor than a moth uses a lamp. They would have found a warm, strangely stained, sickness-haunted spring, and they would have done what humans always did with such places — which is the subject of section 5.

So the honest answer to "by whom and for what": any mining people could have found it; nobody before the twentieth century could have extracted work from it; and the expected archaeological trace of the encounter is not machinery but a sacred site, a healing tradition, a taboo, and an unexplained cluster of illness. Which brings us to the claims, and what they actually are.

4. The genealogy of the claims

Every ancient-nuclear claim this file could find descends from a small family of sources, and naming the family is the fastest way to understand the claims. In 1909, Frederick Soddy — the radiochemist — speculated in his published lectures that earlier civilizations might have known atomic energy and destroyed themselves, a remark made eight years before anyone had built a reactor and three decades before anyone had a reason to think natural ones existed. In 1946, the Soviet writer Alexander Kazantsev published a story proposing that the 1908 Tunguska explosion (the real event that flattened some two thousand square kilometers of Siberian forest) was a nuclear-powered alien spacecraft — the founding fiction of the nuclear-ancient-astronaut genre. From 1943 to 1947, the American pulp editor — pulp being the cheap fiction magazines of the era — Raymond Palmer and a welder named Richard Shaver built a vast mythology in the magazine *Amazing Stories* — an ancient advanced race, abandoned ray machines, good and evil underground peoples — which sold out print runs and drew tens of thousands of letters; its monsters, the "deros," survive today in a famous role-playing game. In 1964, four years before Erich von Däniken, a British Fortean writer — a chronicler of strange phenomena, after the American writer Charles Fort — named W. Raymond Drake published *Gods and Spacemen in the Ancient East*. Von Däniken's *Chariots of the Gods?* followed in 1968 and made the genre global. In 1972, Oklo was announced — and within months, fringe groups were promoting it, in the

words of the Science History Institute's retrospective, as "evidence of antimatter meteorites or crash-landings by ancient aliens in star-cruisers." In 1979, David Davenport published a book claiming Mohenjo-Daro had been destroyed by a nuclear blast two thousand years before Christ; in 1985, Zecharia Sitchin published *The Wars of Gods and Men*, locating an alien nuclear strike in the Sinai in 2024 BC; through the 1990s, David Childress circulated "ancient Indian aircraft technology" compilations; and in the 2000s the whole corpus moved to cable television and streaming. The genealogy matters because it shows these claims are a literary tradition — each generation rewriting the last — rather than a body of evidence that grew.

5. The claims, one by one, with their evidence status

Claim: Oklo was an artificial reactor. The most-cited version attributes doubts to Glenn Seaborg, the Nobel laureate and former head of the US Atomic Energy Commission, on the grounds that reactor water must be "extremely pure" and no natural water qualifies. This file traced the passage to a 1984 young-earth creationist book, Donald Chittick's *The Puzzle of Ancient Man*. No primary Seaborg document contains it; and its physics is wrong on its face — power reactors routinely run with thousands of parts per million of boron in their water, and Oklo's waters carried their poisons comfortably within what its three-and-a-half-percent fuel and meter-sized cores could tolerate. The artificiality claim is documented as evidence of the genre's sourcing habits, no more.

Claim: Mohenjo-Daro was destroyed by a nuclear weapon. The claim rests on a 1966 Soviet popular article by Alexander Gorbovsky — the actual source of the famous "survivors lose their nails and hair" passage that circulates online as a quotation from the Indian epic Mahabharata — and on David Davenport's 1979 book. The archaeology has been examined for decades: George Dales's 1964 study, later forensic re-analyses, and modern isotope work found no destruction layer, no radiological anomaly, and skeletons spanning a thousand years. The "radioactive skeletons" claim traces to no published measurement at all. Documented as a claim; contradicted by the excavated record.

Claim: the Mahabharata describes a nuclear war. The passages circulated — a single projectile "charged with all the power of the Universe," brightness of ten thousand suns, hair and nails falling — exist in the epic in *different scenes*, assembled into one quotation only in twentieth-century fringe compilations. The epic does contain genuinely striking descriptions of weapons and burning cities; poets describe fire well. The symptom list is, as section 3 noted, a real description of radiation sickness — as real as any fire-poem's description of burns. It is worth being exact about the list, because precision is this file's trade. Against the modern medical picture of acute radiation sickness — nausea and vomiting in the first hours, a quiet latent period, then hair loss beginning around the eleventh day after a heavy dose, skin reddening and ulceration, marrow failure over the following weeks — the circulated symptoms decompose honestly: "hair and nails fell out" matches the real

course of epilation; "corpses burned beyond recognition" matches thermal or blast injury of any era; "bright as ten thousand suns" is a flash description any fire-storm poet could write; and "pottery broke without cause" and "birds turned white" have no mechanism in radiation physics at all. About half the list is medically real, a quarter is any-catastrophe imagery, and a quarter is physically impossible — which is exactly what you would expect from a poet composing within a tradition that knew fire, war, and plague, and knew nothing of the nucleus. The claim survives neither as documentation nor as debunking, but as evidence of how convincingly radiation sickness mimics the oldest descriptions of divine punishment — a resemblance that says something true about both.

The claim is documented as a modern collage built on ancient poetry, and its 1986 Soviet popularizer Gorbovsky is named as its author.

Claim: Zecharia Sitchin's 2024 BC Sinai strike. Sitchin's narrative derives from his own re-translations of Sumerian texts, which Assyriologists (specialists in the ancient languages of Mesopotamia) including Michael Heiser have shown to be unsupported — the Akkadian terms he renders as weapons names are common words, and the epic he dates to the third millennium survives in first-millennium manuscripts. His three physical exhibits — radioactive Dead Sea springs, an abandoned-settlement layer, blackened Sinai stones — each have ordinary geological explanations, and, as a careful critical review notes, a detonation of the scale he describes would leave vitrified sand datable by thermoluminescence (a technique that

reads, from the minerals themselves, when they were last heated) and fission-product isotopes in the soil; no search for any of these has ever been published. Documented as a claim with its scholarly critique.

Claim: Tell el-Hammam was destroyed by a cosmic airburst — possibly "the Sodom event." This one is different: it was a peer-reviewed paper, in *Scientific Reports* in 2021, claiming a Tunguska-class airburst — an object that explodes high in the atmosphere rather than striking the ground — destroyed a city of the Bronze Age (the metalworking age between stone and iron) near the Dead Sea around 1650 BC. This file records its full rise and fall, because it is instructive: the claim drew immediate methodological criticism from impact specialists, survived an expression of concern, and was **retracted in April 2025** for errors in analyses, data, and methods, after which its authors republished an expanded version outside peer review. Mainstream and fringe media had by then spent four years calling it, in one magazine's words, evidence that Sodom was destroyed by "an atomic bomb." The destruction layer under the city is real; the cause is undetermined. Documented — including the retraction, which is itself evidence of how the record corrects.

Claim: the Younger Dryas impact was a nuclear catastrophe. The hypothesis that a cosmic impact triggered the Younger Dryas cold snap began life, remarkably, in 2001 under the title "Terrestrial Evidence of a Nuclear Catastrophe in Paleo-Indian Times," in a newsletter of the Center for the Study of the First Americans — its first published form was explicitly

nuclear. It was recast as a comet impact for the 2007 paper in the Proceedings of the US National Academy of Sciences, and after twenty years of dispute, a 2023 comprehensive rebuttal by specialists — one of whom had been the government's airburst physicist — called the evidentiary basis flawed, and the hypothesis remains widely rejected, though its proponents continue to publish. Documented with both sides.

Claim: Libyan Desert Glass is trinitite from an ancient blast. Libyan Desert Glass is real: twenty-nine-million-year-old natural glass spread across the Egyptian-Libyan border, produced by an impact or airburst. The glass that nuclear tests leave behind in the desert sand is called trinitite, and its chemical similarity is exactly why the claim reaches for it. The Libyan glass was genuinely collected and worked in Pharaonic times — the scarab set in Tutankhamun’s pectoral — the chest ornament of his burial regalia — is made of it — and in July 2026 an ancient-astronaut website promoted it as detonation glass matching a thermonuclear strike. The impact literature answers with direct evidence the claim lacks: shock-formed mineral phases and an isotopic age twenty-nine million years deep. The claim is documented as the current live form of the Sinai-nuke story, attaching itself to a real object and misreading it.

Claim: the Baghdad battery and related "ancient electricity." A clay jar with a copper cylinder and iron rod from Parthian-era Iraq, found in 1936, which with acid produces one to two volts. Its function is genuinely undetermined — electroplating, medicine, or simply a storage jar are all proposed — and the artifact was lost in the 2003 war. It is not nuclear, and

this file records it only as the adjacent claim that the nuclear claims borrow credibility from. The fringe habit of collecting vitrified ruins — Scottish hillfort walls, burned Anatolian cities — as blast evidence is answered by ordinary archaeology: intense fires vitrify stone; it happens in sieges and accidents without any nuclear physics.

Claim: young-earth readings of Oklo. A 1981-82 creationist paper series argued Oklo fits a six-thousand-year timescale; a later creationist consortium proposed accelerated nuclear decay to compress the dates. The creationist literature's own summaries concede the two problems this creates — a heat problem with no agreed solution, and a predicted energy budget far below the measured one. Documented, with its internal concessions, as the only remaining route to "Oklo in human time" — and the weakest.

6. The genuine items the claims keep grabbing

Because this file does not throw away evidence with the claims, here, in full, are the real things:

Libyan Desert Glass, truly used by ancients. Impact glass, 29 million years old, worked into jewelry including Tutankhamun's scarab — the one verified case of ancient people collecting and using a material of cosmic violence, as gemstone, not fuel.

Meteoritic iron, truly used by ancients. Tutankhamun's dagger blade, measured in 2016, is meteoritic iron — 10.8 percent nickel — hammered cold, centuries before smelting — the high-heat extraction of metal from ore — existed; the

Egyptians called it "iron of the sky" and valued it above gold. Older still are the Gerzeh beads, five thousand years old. Around the world, a few dozen genuine meteoritic-iron artifacts exist — Indonesia's keris blades mostly mythologize the tradition, but the real examples anchor it. A five-thousand-year-old iron-meteorite crater, Gebel Kamil, still sits in the southwest Egyptian desert — a meteoritic-iron mine the landscape itself keeps.

Mount Wingen, truly lived with. In Australia, a coal seam has been burning underground for about six thousand years — long enough that Aboriginal tradition knows it as Wingen, "fire/burning," and used its mineral exudates — sulfur, alum, red ochre — as medicine. If a shallow heat source had ever lain under an ancient people, this is what the relationship looked like: healing site, sacred geography, gathered minerals. Not engineering.

The Absheron fires, truly worshipped. Near Baku in Azerbaijan, natural gas seeps have burned for millennia — Marco Polo described them; Zoroastrian (the ancient Persian faith that worshipped fire as purity), Hindu, and Sikh pilgrims built the Ateshgah temple around them over centuries, inscriptions in four languages. The modern gas industry lowered the pressure and extinguished the natural flame in the 1880s. Two thousand years of worship produced ritual, architecture, and myth — and not one joule of extracted work.

Jáchymov, truly handled. In the Bohemian silver town of Jáchymov — the origin of the word "dollar" — miners handled pitchblende for four hundred years from 1512, discarded it as

useless black rock, attributed their lethal lung disease to subterranean dwarfs and metal vapors, and never knew what it was. Even handed the ore directly, pre-modern people could neither sense nor use it. That is the real-world data point that answers the harnessing question: not lack of opportunity — lack of any possible relationship.

And the one case where life genuinely harnessed the phenomenon: the cyanobacteria. Their oxygen dissolved the uranium, the organic matter trapped it, and the whole reactor story exists because biology built the ore. The only engineers of the Oklo reactors were algae — which, as Lovelock wrote, ran them for a million years before fire existed.

7. The suspects roster — who the claims name

For completeness, the named builders across the literature: the Anunnaki of Sitchin's translations; the Atlanteans and Lemurians of the Shaver-Palmer mythos; "pre-flood" high-technology humanity of the creationist-crypto literature; the Rama Empire of the vimana compilations (vimanas being the flying vehicles of the ancient Indian texts); the extraterrestrials of von Däniken and Kazantsev; and the unnamed "advanced civilization" of the viral posts. What is striking, and this file says so plainly, is that every one of these suspects is invoked to explain a phenomenon whose actual explanation — decay-law enrichment, groundwater moderation, clay preservation, a biosphere that built the ore — is better documented than any suspect has ever been.

8. Verdict of Part IX

One legitimate descendant of the fringe questions deserves the last word before the verdict, because scientists themselves ask it. In the modern search for technosignatures — signs of past technology on other worlds or our own — researchers including Paul Davies have pointed out that Oklo is the calibration case: an isotopic anomaly that *looked* artificial and proved natural. The discipline's rule of thumb is Davies's: had Oklo been dated younger, there would not have been time for a natural reactor to exist, and the anomaly would have demanded a very different explanation. The isotopes are the same for the believer and the skeptic; only the inference discipline differs — which is the whole lesson of this part.

No evidence exists that any person, at any time before 1972, found, used, or understood a natural nuclear reactor. The physics forbids the timing; the engineering forbids the use; the archaeology is absent — the reactor ground was first disturbed by COMUF's blasting in 1970, and no trace of any earlier digging exists anywhere in the district. Every positive claim traces to a documented literary genealogy beginning with Soddy's 1909 speculation, and every positive claim fails on evidence this part has laid out. The claims are kept, in full, as evidence — of a century's longing for deep-time ancestors and deep-time warnings, which is itself one of the most human things this file contains.

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PART X — THE DEEP-TIME CIRCLE

1. Saving the last reactor

By the early 1990s the mining had done what mining does. Of the sixteen reactor zones, every one in the Oklo-Okélobondo deposits was either mined away or left unreachable in the flooded workings; a section of Zone 2 had been pinned to the pit wall in 1976, and even that survives only as faces and archived cores. One reactor remained intact: Bangombé, twelve meters down, too shallow to have been worth the cost of digging.

In May 1997, François Gauthier-Lafaye and his European colleagues published a short plea in the journal *Nature* — a correspondence, not a paper — that deserves to be quoted in full, because it worked: "The last known natural fission reactor on Earth is likely to be mined this year... All the reactors except one are located in the most important uranium deposit of Gabon's Franceville basin, at Oklo... This deposit will be completely mined out soon, in 1998. Future work on these reactors will therefore have to rely on previously collected samples, many of which are poorly documented and are out of their geological context... We propose that this unique, scientifically important deposit be preserved for present and future research. This deposit is no less unique, and certainly more irreplaceable, than the most valued specimens from the Moon and Mars."

Gabon listened. On June 2, 2017, the Gabonese government classified the Bangombé reactor and the nearby Moulendé fossil beds as protected historical ensembles, placing them under the

surveillance of the national parks agency; access for science requires a research permit from the parks authority and, for any excavation, authorization from the Minister of Culture. In January 2022, Gabon submitted both sites — now formally paired, the two-billion-year-old fossils of the Francevillian biota and the two-billion-year-old reactor — to the United Nations' world-heritage program, UNESCO, and its tentative World Heritage List, where they stand today, awaiting nomination. Gabon issued a postage stamp honoring the reactors in 1976, part of a four-value set catalogued as Michel 578-581; the discovery had become, properly, a point of national pride. And in March 2018, two half-split drill cores from the reactor zones were presented to the Natural History Museum in Vienna — funded by Orano and the CEA, with the IAEA assisting — where they sit in the permanent collection at about forty microsieverts per hour at five centimeters, roughly the dose rate of a long-haul flight, exhibited beside a world map of natural radioactivity. The curator, Ludovic Ferrière, tells visitors: rocks are like books; the cover gives you the basics, but you have to open them.

2. The full circle: what we teach the future

Here is the strangest lesson Oklo teaches, and it took this file until now to say it plainly. For fifty years, the world's nuclear waste programs have faced a problem no engineering can solve by itself: how to tell people a hundred thousand years from now — people who may not speak any language we know, who may not be us, biologically — that a hole in the ground is dangerous and must be left alone. Finland is drilling Onkalo, a repository in granite designed to be sealed and forgotten for a hundred millennia; the filmmaker Michael Madsen stood inside it and addressed the future intruders on camera: "This place is not a place of honor… you are heading toward a place where you should never go."

The designers of such markers studied everything — forbidding landscapes of thorns, fields of spikes, engraved warnings in six languages, even Thomas Sebeok's half-serious proposal of an "atomic priesthood," an order of specialists whose ritual duty would be to remember. And through all of it runs one precedent, used explicitly in the disposal literature: a mine in Gabon, where nature buried nuclear waste, and where the future people — us — dug into it, misread it at first as a mystery, and then understood it. Oklo is the demonstration, etched in isotopes, that a dangerous place *can* be read correctly across deep time by whoever comes next. The waste community's own documents put the limits honestly — Oklo is a good presumption, not a demonstration — but the precedent is real, and it cuts both ways: it proves containment works, and it proves that even the

best-sealed story can be opened by a curious species with mass spectrometers.

Weinberg's old worry belongs here too. He coined the word "trans-science" for questions science cannot settle — the effects of insults too small to measure — and he warned that public fear of such small insults resembles, in his phrase, the fear of witches in the later Middle Ages. The waste markers, the priesthoods, the spikes in the desert, and the flooded mine in Gabon are all one conversation: how a clever, anxious species talks across the only timescale that finally matters. Oklo is the one voice in that conversation that speaks from two billion years ago, and it says: it held.

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PART XI — TODAY

1. The site today

The Mounana mill is dismantled; the pits and galleries are flooded; the district's last output ended in 1999 and the rehabilitation was completed in 2004 with European Union money, seven million euros of it, the tailings dammed and covered under a monitoring program the government runs. Mounana the town — 6,500 people at the mine's height — is pivoting to manganese processing, and its official development plan for 2021-2026 does not mention uranium; the radioactive century is being quietly folded into the past. Gabon's known uranium resources, some 4,800 tons, sit uneconomic in the Red Book. Orano's Gabonese subsidiary, twenty-seven years after the last ore, still exists, with a new president named in late 2026. And at Bangombé, under the protection of the 2017 decree, the last reactor rests twelve meters down, watched by the parks agency, waiting for the next scientist with a permit.

2. The science today

The research did not end when the samples ran out; it changed instruments. In 2018, the Naval Research Laboratory's isotope-imaging team read single metal grains and found cesium and barium captured within five years of shutdown. In 2021, the same group timed the cesium flush-out to about three years and confirmed nearly complete technetium retention in the metal alloys. In 2023, the Strasbourg-Algerian team modeled the clay's role in the reactors' birth, and in February 2026 Takahiro Hidaka's group — now joined by a Gabonese researcher, Evaldie Durastanti Rabnga Mombo — published a call to arms: most Oklo isotope data are more than twenty-five years old, and modern mass spectrometers could still teach us new things. The 2022 fiftieth-anniversary symposium in Strasbourg gathered the living lineage — Gauthier-Lafaye, Dozol, Nimal, Gall, Bentridi, Meshik, Hidaka — and its proceedings record the standing assessment: the discovery answered its first questions within months; some of its finest, including the full thermal history and the small-zone ignition problem, are still open.

3. The company called Oklo

In 2013, two MIT-trained nuclear engineers, Jacob DeWitte and Caroline Cochran, founded a reactor startup in Santa Clara and named it, openly and with the naming statement on record, after the Gabon phenomenon: "derived from Oklo, a region in the country of Gabon, Africa, where self-sustaining nuclear fission reactions occurred approximately 1.7 billion years ago." Backed early by Sam Altman, the company went public in May 2024 through a SPAC — a shell company created for the purpose, the fashionable shortcut to the stock market — and began trading on the New York Stock Exchange under the four-letter name OKLO. Its design — a sodium-cooled fast reactor descended from America's 1960s Experimental Breeder Reactor II — has had a hard road and a fast one: the Nuclear Regulatory Commission declined its first license application in 2022; the Energy Department's reactor-pilot program admitted it; and in August 2026, its zero-power test assembly called Groves achieved criticality in Texas — the same milestone the Oklo cores crossed 1.95 billion years earlier: a sustained chain reaction, lit for the first time — the fastest privately sited, privately built reactor to reach that milestone, built in 229 days. Its third-quarter 2026 results, published on November 6, 2026, reported revenue of 1.86 million dollars, its first real sales, against the heavy spending of a first-of-a-kind builder; its market value has swung from a peak above ten billion dollars down roughly half as the market argues with itself about whether regulatory de-risking and technical de-risking are the same

thing — a debate its own filings join candidly. In September 2026 it lost a federal complaint over a stalled grid-interconnection application in Virginia — regulators found the application incomplete — a reminder, in its own small way, that nature's reactor never needed a permit, and ours always will. This file records the company, its namesake, and its tribulations as the afterword they are: humanity is still trying to build what Gabon built by accident, and the attempt, whatever its fate, keeps the old discovery in the news. In November 2026 the Energy Department's ARPA-E program — its arm for funding long-shot energy technologies — named it one of eight funded developers of recycled nuclear fuel — including, alongside it, the established giant Westinghouse — a small coda for a company named after the one reactor that never needed fuel fabrication at all.

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EPILOGUE: WHAT THE ROCK SAYS

We close the way the investigators closed their anniversary symposium — with what is known, what is not, and the words the file has carried all along.

Known, beyond reasonable dispute: two billion years ago, in a basin in what is now Gabon, groundwater-moderated nuclear reactors ran at ten to a hundred kilowatts for tens to hundreds of thousands of years each, cycling roughly half an hour on and two and a half hours off, breeding and burning plutonium, controlling themselves with nothing but water, clay, and the Doppler effect, and then shut down; their actinides and most of their fission products stayed within meters of home for two billion years; and the laws of physics, checked against their samarium, have not changed by more than a part in a hundred million over that whole span.

Not known, openly: the exact operating temperatures; how the smallest zones ignited; the whole-field energy budget to better than a factor of two; the full story of the people who mined the ore, which the courts and the epidemiologists have not finished telling; and whether anything like Oklo will ever be found again — the searches of four continents say no, so far, and the physics says the window has closed forever.

The words we carried: Jules Horowitz, in his foreword to Naudet's book, called it a miracle that the evidence survived the upheavals of two billion years to be found by watchful analysts. George Cowan said that a Precambrian physicist would have found it almost easy to build a reactor, and that in retrospect it

seems inevitable it happened accidentally. Shlyakhter said the designers of nuclear reactors were not innovators but unwitting imitators of nature. Lovelock said a community of modest bacteria built reactors that ran for a million years. Eklund, the IAEA's director general, stood in the pit in 1975 and asked that the fossil reactor not suffer the same fate as Fermi's Chicago pile, which was demolished without a monument. Gauthier-Lafaye said the last reactor was more irreplaceable than anything brought back from the Moon. Meshik, who read the pulsing clock in a mineral grain, said simply that nature is the first genius. The Swiss scientific community's own journal remembered how the discovery sounded in the mid-1970s: whispered in university corridors in awe by some, in disbelief or even disdain by many — "Fantastic! Impossible!" And the file's own rule, one more time: every one of these people, and every claim examined in Part IX, and every error catalogued in Appendix B, is evidence — and the evidence, like the uranium, has a fingerprint that honest weighing can read.

Thank you for reading the whole file. It was a pleasure to explain it.

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APPENDIX A — THE MASTER TABLES

A.1 The reactor zones (canonical count: 16 = 15 Oklo-Okélobondo + 1 Bangombé, Gauthier-Lafaye 2002)

Zone	Found	Where	Depth	Notes
1–6	1972–1974	Northern Oklo pit	~100–260 m	Zones 1, 2, 5-6 on the fold hinge; 3-4 deeper, found by drilling
7–9	1978	Southern Oklo pit	~150 m	Rich in organic matter; RZ9 the best-modeled small core
10	1982	Oklo, underground	~400 m	"Cold" reactor: minerals record 100–120 °C
13	1984	Oklo, near dolerite dyke	deep	Highest fluence; most depleted U known (0.366%)
15	early	Oklo	pit	Mined before EC sampling
16	1982	Oklo	deep	Apatites carry Pu + fission products
OK84 (Okélobondo)	1982, drilling	South end, 1.5–2 kilometers south	about 450 meters	Least weathered; the repository analogue
11, 12, 14	—	—	—	Numbered candidates never confirmed as reactors
Bangombé	1982 (deposit); 1986 (reactor, drilling)	30 kilometers south of Oklo	12 meters	The only survivor; preserved

A.2 Per-zone operating parameters (best measurements; each carries its own paper)

Zone	U-235 burned	Duration	Fluence (n/cm^2)	Fission shares 2 35U/238U/239Pu
RZ2 (largest)	~1,800 kg (~2 tons)	~850,000 yr	~8.7×10^20	~95/2/3 %
RZ10	~650 kg (~1,400 lb)	~160,000 yr	~5.6×10^20	~91/5/3 %
RZ9	—	~220,000 yr	~3.6×10^20	~88/8/4 %
RZ13	—	shortest, highest fluence	~1.2×10^21	~75/18/7 % (most fast fissions)
Bangombé	—	~100,000–800,000 yr	~3×10^20	~92/4/4 %

Whole field: ~5–6 tons (10,000 to 12,000 pounds) of uranium-235 fissioned; ~10,000–16,000 megawatt-years (~ 420 petajoules ~ 100 megatons of TNT); ~1.5–2.5 tons (3,000 to 5,500 pounds) of plutonium-239 bred; mean power 10–100 kilowatts per zone; cycle roughly half an hour on, two and a half hours off; temperature ~150–450 °C (best band 200–350).

A.3 Production record (authorities named in Part II)

- Red Book lifetime total: 25,403 tons of uranium (the short ton of 2,000 pounds is meant throughout this book). Variants: 25,635 (UDEPO 1962-99); 27,872 (OECD-IAEA 2004); 28,492 (2002 per-mine table); 26,000/26,660 (Areva communications).
- Per mine (2002 OECD table), in tons: Mounana 5,759; Oklo 15,342; Okélobondo 3,500; Boyindzi 2,691; Mikouloungou 1,200.
- Peak about 1,250 tons per year (1974–79). Mined reactor-zone ore: 94 tons (isotopically depleted).

A.4 Fission-product inventory, corrected

Per zone — about ten pounds of uranium-235 fissioned, at six percent yields — the masses run: Tc-99 ~ 280 g; Cs-135 ~ 290 g; Zr-93 ~ 290 g; Cs-137 ~ 280 g; Sr-90 ~ 265 g; I-129 ~ 41 g; Kr-85 ~ 12 g. (An ounce is about 28 grams, so even the largest entry is a letter-stamp's worth of material. Corrects the Wikipedia table overstating by ~450×; see Appendix B, item 24.)

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APPENDIX B — THE FACT-CHECK FILE

Every error or inconsistency found in the published record during the research, numbered, with its correction or standing note.

1. IAEA outreach credits Perrin with the discovery analysis. Primary record: he presented the Academy notes; the analysts were Bouzigues (Pierrelatte) and the Cadarache team. Giraud announced for the CEA.
2. The 1972 Academy notes are in Comptes Rendus vol. 275 (pp. 1731, 1847, 2291); a major 2014 review printed "237" — propagated error.
3. Same review printed Petrov's Soviet Physics Uspekhi paper with year 1997; it is 1977 (vol. 20, p. 937; original Russian Uspekhi Fiz. Nauk 123:473).
4. Same review attributed the Interdisciplinary Science Reviews piece to "Irvine & Naudet"; the author was R. Audet (with Naudet).
5. The 1975 symposium venue: Libreville (official); Naudet's preview said Franceville. Discrepancy lives in the primary record.
6. U-235 natural abundance: 0.7202 wt% = 0.7110 at%. Sources mix these constantly; standing rule — always state the unit.
7. Popular age/enrichment pairings are internally inconsistent: "1.7 Ga at 3.1%", "1.8 Ga at 1.8%", "2 Ga at 5%" (analog report). Canonical: 1,950±40 Ma at ~3.5 wt%.

8. IAEA's own World Uranium Geology document prints "1.8% at 1800 Ma" — an IAEA-popular-source error.
9. A French parliamentary report (Bataille-Galley) says Bangombé ran "500 years" — wrong by ~200×; and "15 Oklo reactors" count variant.
10. Wikipedia's natural-reactor article: per-zone FP masses overstated ~450× (129 kg — 284 pounds — of Tc-99 per zone); whole-field 420 petajoules printed as per-reactor. Corrected in Appendix A.4.
11. en.wiki "Nuclear reactor" gives 1.7 Ga/3.1% — same error class.
12. "150 million years" operating duration circulates in popular media; geochemical truth is hundreds of thousands of years. (Now corrected by the best current popular writing.)
13. The Seaborg "pure water" quotation: no primary source; traced to Donald Chittick, The Puzzle of Ancient Man (Creation Compass, 1984). Physics doubly wrong (PWRs run thousands of ppm boron; Oklo's poisons were ppm-scale).
14. The "Mahabharata nuclear passage" as circulated is a modern collage (Childress/Burrows compilations); its sickness description traces to Gorbovsky, Sputnik magazine, Sept 1986, p. 137.
15. The Oppenheimer "Rama Empire/vimana" quotations are fabricated; his genuine quote is the Bhagavad Gita line (1945).
16. The "50× radioactive skeletons at Mohenjo-Daro" figure: no published measurement; source untraceable.
17. Tsinghua-university-textbook passage ("thirteen reactors... comparable to the modern nuclear reactor in power"): wrong

count and grossly inflated power; unverified against the actual textbook.
18. Clickbait variants: "operated for about 150 years at 150 kW" (garbled from 150,000 yr / <100 kW); "Kuroda predicted it in 1952" (1956).
19. en.wiki List of uranium projects dates Oklo's start to 1956 (that's Mounana).
20. An ACS book chapter dates Kuroda's prediction to 1956 at "Oklo, Gabon" — anachronism (Oklo unknown until 1968; phenomenon 1972).
21. A University of Arkansas community blog claims a natural reactor "in Wisconsin" — bogus; no such literature.
22. Portuguese popular timelines: "mais de dois mil anos" (two thousand years) for two billion.
23. IntechOpen 2020 chapter: 2.5 Ga age, 200-Myr duration, rift-timing conflation — three errors in one passage.
24. (Consolidated with item 10 — the FP/energy table.)
25. A 2026 peer-reviewed public-health paper credits "Perrin and his colleagues" with the discovery — the credit-chain error persists in print.
26. The Madinda 2026 paper uses generative AI (ChatGPT) in drafting — disclosed by the authors; recorded for reader context.
27. The Groopman 2021 citation exists in two printings: Applied Geochemistry 124:104861 vs 131:105047; the publisher record supports 131.
28. "Pierrelatte, Gabon" conflation in fringe literature — the French plant and the Gabon site merged into one fictitious

place.
29. Gabon-stamp catalogues list a 1975 design-year entry; the documented issue is 15 December 1976, Michel 578-581 (four values, 60 CFA francs, the local currency — about a quarter at the time).
30. Hungarian atomic-history page says "Oklo 15 intact and viewable" — it describes the pinned Zone 2 wall section; RZ15 was mined out.
31. Yahoo-finance-style data conflicts on Oklo Inc. YTD return (+70% vs −50%); aggregator earnings dates (Nov 10/18 vs official Nov 6); a transcription error that turned a 1.19-billion-dollar at-the-market share sale into “1.19 million”. SEC filings are the authority.
32. The zone count: 14/15/16/17 variants — resolved: 16 total per Gauthier-Lafaye 2002 (15 Oklo-Okélobondo + Bangombé).
33. Production totals: 25,403/25,635/26,000/26,660/27,872/28,492 tons — resolved: Red Book 25,403 as headline; variants = accounting scope.
34. Discovery date: May / 2 June / 14 June / June 1972 across primary sources — Dozol's first-person 14 June weighted.
35. "150-million-year" and "1.8%" figures (IAEA popular documents) — flagged; do not propagate.
36. Mossman et al. 2008 prints the reactor age as "1968±50 Ma" and "up to 1 Ma" durations — team variants; canonical 1950±40 Ma and per-zone 20k–850k yr.
37. The Ruffenach 1976 EPSL paper circulates with French and English titles — same paper.

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APPENDIX C — THE BIBLIOGRAPHY

Organized for the general reader, with the full citation record noted as living in the evidence files. Primary sources first within each group.

C.1 The proceedings and syntheses

- *The Oklo Phenomenon*, IAEA Proceedings Series STI/PUB/405, Libreville, 23-27 June 1975 (646 pp.). The founding volume.
- *Natural Fission Reactors*, IAEA STI/PUB/475, Paris, 19-21 December 1977 (754 pp.). The second conference; the waste-analog framing begins.
- R. Naudet, *Oklo: des réacteurs nucléaires fossiles — Étude physique*, Éditions Eyrolles, Paris, 1991. The standard physics monograph (with Jules Horowitz's foreword).
- F. Gauthier-Lafaye, *Les gisements d'uranium du Gabon et les réacteurs d'Oklo*, Mémoires Scientifiques Géologiques 78, Strasbourg, 1986. The definitive geology thesis (open access via Persée).
- F. Gauthier-Lafaye, "Oklo," C.R. Physique 3:839-849, 2002; and Gauthier-Lafaye et al., C.R. Geoscience 343 (2011) and 2016 — the team's standing summaries.
- The fiftieth-anniversary set: Radiation Protection Dosimetry 199(18), November 2023 (Gauthier-Lafaye et al. ncad044; Dozol ncad014; Nimal ncad043; Bentridi et al. ncad048; Gall ncad034; Hidaka et al. ncad015), from the Strasbourg

ICHLERA-2022 symposium.

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APPENDIX D — GLOSSARY AND THE PEDIGREE KEY

D.1 Words this file could not avoid

Uranium. A heavy metal element — nearly as dense as gold — whose unstable atoms can be split to release energy; the fuel of reactors and of this whole story. **Atom and element.** Everything is made of atoms; an element is one kind of atom, defined by the number of protons — the positively charged particles at its center — in its nucleus. **Neutron.** An uncharged nuclear particle; the projectile that can split a uranium-235 nucleus, and the currency of every chain reaction. **Radioactivity.** The spontaneous transformation of unstable atoms, releasing energy and particles on fixed schedules (half-lives) that nothing can hurry or slow. **Fission.** The splitting of a heavy nucleus into two lighter ones, releasing energy and two to three new neutrons. **Isotope.** A flavor of an element. Uranium's two important flavors, uranium-235 and uranium-238, are the same element chemically but have different atomic weights and different nuclear hearts: 235 can sustain a chain reaction; 238 mostly cannot.

Half-life. The time in which half of a batch of an isotope decays away. Uranium-235's half-life is 704 million years; uranium-238's is 4.47 billion. This is why the uranium-235 share was higher in the past — the fast-decaying one used to be more common.

Chain reaction. One neutron splits a nucleus, which releases more neutrons, which split more nuclei. If the bookkeeping holds — on average one neutron from each split causing another split — the reaction sustains itself. That condition is called *criticality*, and a system that achieves it is said to be *critical*.

Moderator. The material that slows neutrons down so they split uranium-235 efficiently. Ordinary water works; so did the groundwater in the Oklo rock, two billion years ago.

Neutron poison. Anything that eats neutrons without splitting — the boron in control rods, or, at Oklo, rare earths like samarium and gadolinium. *Burnable poisons* are ones that get used up as they absorb, smoothing a reactor's life.

Void coefficient. The response of a water-moderated reactor when its water boils away: the reaction slows and stops. Negative void coefficient means the reactor shuts itself down when it gets too hot — Oklo's governing safety system, built in.

Doppler broadening. When fuel heats, its atoms jiggle faster, smearing the energies at which it absorbs neutrons — one more automatic brake, present in every modern reactor and in Oklo.

Fluence. The total number of neutrons that passed through a square centimeter over a reactor's life — the integrated "neutron dose." Oklo fluences ran up to about a million billion neutrons per square centimeter (10^21).

Fission products. The fragments left when a nucleus splits — dozens of new elements, from krypton and xenon through technetium and ruthenium to cesium and the rare earths. They are the waste, and at Oklo, the witnesses.

Technetium-99. A fission product with a 210,000-year half-life, important because it is mobile in water and long-lived; at Oklo it decayed into ruthenium-99, whose extra abundance is one of the fingerprints. Half-life means half of it decays in that time, so measurable traces survive far longer.

Pitchblende and uraninite. Two names for essentially the same mineral: uranium oxide ore. Geologists say *uraninite* for the crystalline form, *pitchblende* for the massive ore; at Oklo, the reactor cores are uraninite.

Argile de pile. "Pile clay" — the zoned clay jacket each Oklo reactor grew around itself as it ran, part sarcophagus, part heat record.

Great Oxidation Event. The planet-wide rise of oxygen beginning around 2.4 billion years ago, produced by photosynthetic microbes. It made uranium water-soluble and therefore minable by nature. Without it, no Oklo.

Proterozoic / Precambrian. Names for the eon before complex life: from 2.5 billion years ago to 539 million years ago. The Oklo rocks are early Proterozoic.

Mass spectrometer. The instrument that weighs atoms one isotope at a time — the machine that read Oklo's fingerprints. The 1972 discovery rode on one at Pierrelatte; the modern versions are a hundred times more sensitive.

INES. The International Nuclear and Radiological Event Scale, the standard way to rate nuclear accidents. Chernobyl and Fukushima are level 7. Mounana is unrated — a fact the 2026 public-health paper highlighted in its comparison table, and which this file records as a framing choice, not a fact of

physics.

ALARA. "As low as reasonably achievable" — the regulatory principle behind the Mounana cleanup target of keeping added public dose under one millisievert per year.

Millisievert (mSv). The unit of radiation dose. Background nature gives you two to three per year. Symptoms of acute radiation sickness appear above about 500 in a short burst; a sievert is a thousand millisieverts. **Becquerel (Bq).** The unit of radioactivity itself: one atomic decay per second. The river measurements at Mounana were made in these units. **Curie and picocurie.** The older units still common in the literature: one curie is 37 billion decays per second, so a picocurie is about one decay every three seconds.

Monte Carlo. The workhorse method of modern reactor physics: computer models that follow individual neutrons through simulated material, millions of times over, and average the outcomes. **HALEU.** High-assay low-enriched uranium — fuel enriched to just under 20 percent, the stuff the modern company named Oklo plans to burn, and the same neighborhood of enrichment nature provided at Oklo two billion years ago.

D.2 The pedigree key

Throughout the research behind this file, every source carried a letter saying what kind of thing it was. For transparency, here is the key:

- **[S] — Source-grade primary.** The scientific literature, official agency records, court filings, corporate financial statements, conference proceedings, and declassified

government documents.

- **[A] — Authority-grade secondary.** Wikipedia and reference works used to locate primaries, plus reputable journalism.
- **[NA] — Named/attributed.** Individual experts' pages, conference programs, institutional repositories — reliable but informal.
- **[B] — Popular.** Books, magazines, podcasts, and general press — used for color and quotes, never for load-bearing numbers.
- **[C] — Claim literature.** Fringe, advocacy, and promotional material — documented as evidence of the claim, never as evidence of the fact.
- **[D] — Derived.** Calculations this file performed from [S]-grade data (for example, the energy cross-checks in Appendix A).

When sources of different grades disagreed, the higher grade governed; when two [S] sources disagreed, both are shown, as with the discovery date, the dolerite ages, and every other honest dispute in this file.

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End of the file. Compiled 2026-11-07 from an 86-dive evidence record; stored in full, uncompressed, at the locations noted in Appendix C. Everything checkable is cited; everything disputed is shown; everything unknown is named.